

A Mini Project Report
on
STUDENT PERFORMANCE PREDICTING SYSTEM USING
RANDOM FOREST ALGORITHM
BACHELOR OF ENGINEERING
in
CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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Methodist College of Engineering and Technology,
King Koti, Abids, Hyderabad-500001 2025-2026

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DECLARATION BY THE CANDIDATES

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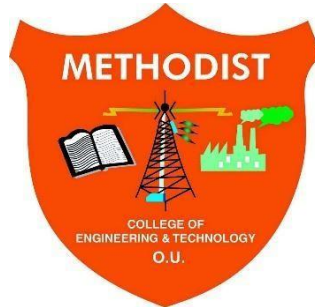
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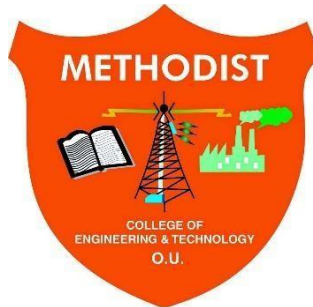
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INTERNAL

EXTERNAL

HOD

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Vision & Mission

VISION

To become a leader in providing Computer Science and Engineering education with emphasis on knowledge and innovation.

MISSION

M1: To offer flexible programs of study with collaborations to suit industry needs

M2: To provide quality education and training through novel pedagogical practices

M3: To Expedite high performance of excellence in teaching, research and innovations.

M4: To impart moral, ethical valued education with social responsibility.

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- PEO3:** Promote collaborative learning and spirit of team work through multidisciplinary projects
- PEO4:** Engage in life-long learning and develop entrepreneurial skills.

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PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological chan

ABSTRACT

The Student Performance Predicting System Using Random Forest Algorithm is a web-based machine learning application developed to predict students' academic performance and provide personalized recommendations for academic improvement. The system analyze multiple factors including study hours, attendance percentage, sleep duration, social media usage, previous academic scores, participation in extracurricular activities, part-time employment status, internet accessibility, and parental education level. Based on these inputs, the system classifies students into performance categories such as High, Medium, and Low using a trained Random Forest Classifier.

The application calculates prediction confidence scores, determines academic risk levels, and generates personalized recommendations aimed at improving study habits, learning efficiency, attendance, and overall academic achievement. The system is implemented using Python, Flask, Scikit-Learn, SQLite, HTML, CSS, JavaScript, and Chart.js. It incorporates role-based authentication for Students, Teachers, and Administrators, along with prediction history management, dashboard analytics, graphical visualizations, and performance monitoring features.

The proposed system supports data-driven educational decision-making by enabling early identification of students requiring academic assistance, facilitating continuous performance monitoring, and providing actionable insights to enhance student success. By integrating machine learning techniques with web-based analytics, the system offers an effective and scalable solution for modern educational institutions.

TABLE OF CONTENTS

S. No	Title	Page No.
	Certificate	I - V
	Acknowledgement	VI
	Abstract	IX
1	Introduction	1-2
1.1	Problem Statement	3
1.2	Objective	4
1.3	Scope	4
1.4	Organization of Report	5
2	Literature Survey	6-7
3	System Analysis	8
3.1	Existing System	8
3.1.1	Drawbacks	8
3.2	Proposed System	8
3.2.1	Advantages	9
3.3	Applications	9
4	Software Requirement Specifications	10
4.1	Overview	10
4.2	Hardware Requirements	10
4.3	Software Requirements	10
4.4	Functional Requirements	11
4.5	Non Functional Requirements	11
5	System Design	12
5.1	Input Design	12
5.2	Processing Design	13
5.3	Output Design	14
5.4	System Architecture	14
5.5	Workflow	15

S. No	Title	Page No.
5.6	UML Diagrams	16-24
5.7	Data Flow Diagrams	25-26
6	Implementation	27
6.1	Technologies Used	28
6.2	System Architecture	29
6.3	Module Description	30-33
6.4	Working Procedure	34
7	Testing	35
7.1	Test Cases	35
7.2	Testing Methods	36
7.3	Results of Testing	36
8	Results	37
8.1	Present Results Obtained	37
8.2	Screenshots	38-50
8.3	Performance Analysis	51
8.4	Interpretation Of Results	52
9	Conclusion	53
9.1	Future Enhancements	54
10	References	55-56
11	Appendix	57
	Project SDG Mapping	58-60
	Quality Analysis	61

LIST OF FIGURES

Figure No.	Title	Page No.
5.6.1	Use Case Diagram	15
5.6.2	Class Diagram	17
5.6.3	Sequence Diagram	18
5.6.4	Activity Diagram	20
5.6.5	Component Diagram	21
5.6.6	Deployment Diagram	22
5.7.1	Level 0 DFD	23
5.7.2	Level 1 DFD	24
6.2.1	System Architecture	27
6.4.1	Working Procedure	32
8.2.1.1	Home page	36
8.2.1.2	Role Selection Page	36
8.2.2.1	Student Login Page	37 - 38
8.2.2.2	Performance Analytics	38
8.2.2.3	Daily Activity Tracker	39
8.2.2.4	Student Analytics Dashboard	39 -40
8.2.2.5	Previous Score input	41
8.2.2.6	Student Information	41
8.2.2.7	Recommendation Dashboard	42 - 43
8.2.3.1	Teacher Login Page	44
8.2.3.2	Teacher Analytics Dashboard	44 - 46
8.2.4.1	Admin Login Page	47
8.9	Admin Dashboard	47 - 48

LIST OF TABLES

Table No.	Title	Page No.
1.4	Organization Of The Report	5
2.1	Literature Survey	7
4.2	Hardware Requirements	10
4.3	Software Requirements	10
4.4	Functional Requirements	11
4.5	Non-Functional Requirements	11
5.1	Input Design Fields	12
6.1	Tools And Technologies Used	28
7.1	Test Cases	35
8.4	Interpretation Of Results	52

CHAPTER 1

INTRODUCTION

Education plays a vital role in personal, social, and economic development. Academic performance is one of the most important indicators used to evaluate a student's learning progress and overall educational achievement. Student performance is influenced by a combination of academic, behavioural, social, and lifestyle-related factors such as study habits, attendance, sleep patterns, social media usage, parental education, and participation in extracurricular activities. Identifying students who may face academic difficulties at an early stage is essential for providing timely support and improving educational outcomes.

Traditional methods of evaluating student performance primarily rely on examination results and periodic assessments. While these methods provide useful information, they often fail to consider the multiple factors that contribute to a student's academic success. As educational institutions continue to generate large amounts of student-related data, there is an increasing need for intelligent systems capable of analyzing this data and extracting meaningful insights that can support academic decision-making.

Recent advancements in Artificial Intelligence and Machine Learning have created new opportunities for educational data analysis. Machine learning algorithms can identify hidden patterns, relationships, and trends within large datasets, enabling accurate prediction of student performance. These predictive models help educational institutions identify students who may be at academic risk and provide proactive interventions before performance declines significantly.

The Student Performance Predicting System Using Random Forest Algorithm is a web-based machine learning application developed to predict students' academic performance based on various academic and lifestyle-related attributes. The system utilizes a Random Forest Classifier, a powerful ensemble learning algorithm known for its high accuracy, robustness, and ability to handle complex datasets. By analyzing factors such as study hours, attendance percentage, sleep duration, social media usage, previous academic scores, extracurricular participation, parental education level, and internet accessibility, the system classifies students into performance categories such as High, Medium, and Low.

In addition to prediction, the system provides personalized recommendations aimed at improving academic performance. Based on the identified strengths and weaknesses of each student, the application generates actionable suggestions related to study habits, attendance improvement, time management, and learning strategies. The system also calculates confidence scores and risk levels, enabling users to better understand the reliability and significance of the prediction results.

The proposed system is implemented using Python, Flask, Scikit-Learn, SQLite, HTML, CSS, JavaScript, and Chart.js. It includes role-based access for Students, Teachers, and Administrators, along with dashboards that provide visual analytics, prediction history, and performance monitoring capabilities. Through the integration of machine learning, recommendation systems, and web technologies, the application serves as an intelligent educational support tool that enhances student success and promotes data-driven decision-making within educational institutions.

1.1 PROBLEM STATEMENT

Educational institutions continuously generate large amounts of student-related data, including academic records, attendance, study habits, and behavioural information. However, traditional methods of evaluating student performance primarily rely on examination results and manual assessment, which may not effectively identify students who are at risk of poor academic performance at an early stage. Such approaches often fail to consider multiple factors that influence student achievement, including lifestyle habits, social behaviour, sleep patterns, and learning practices.

The absence of an intelligent prediction mechanism makes it difficult for educators to provide timely academic support and personalized guidance to students who require intervention. As a result, students may experience declining performance, reduced motivation, and increased academic risk without receiving appropriate assistance.

Therefore, there is a need for an automated and data-driven system that can analyze academic, behavioural, and lifestyle-related factors to accurately predict student performance and identify potential academic risks. Such a system should not only predict performance levels but also generate personalized recommendations that help students improve their study habits, learning efficiency, attendance, and overall academic achievement.

The proposed Student Performance Predicting System Using Random Forest Algorithm addresses this problem by utilizing machine learning techniques to predict student performance, assess risk levels, provide personalized recommendations, and support data-driven decision-making for students, teachers, and educational institutions.

1.2 OBJECTIVE

The main objectives of this project are:

- To predict student performance using machine learning.
- To classify students into High, Medium, and Low performance categories.
- To provide personalized academic recommendations.
- To design a secure web application with role-based access.
- To store prediction history in a database.
- To provide dashboards for students, teachers, and administrators.
- To generate charts and analytics for easy interpretation.
- To support academic planning and early intervention.

1.3 SCOPE

The scope of the project includes performance prediction, recommendation generation, user authentication, dashboard analysis, and prediction history tracking. The system can be used in schools, colleges, and coaching institutions for academic monitoring and early identification of weak students.

The project can also be extended in the future with mobile access, AI chatbot support, cloud deployment, and real-time academic database integration.

1.4 ORGANIZATION OF THE REPORT

This report is organized into multiple chapters to systematically present the development of the Student Performance Predicting System Using Random Forest.

Chapter No.	Chapter Title	Description
1	Introduction	Overview of the project including problem statement, objectives, scope, and report structure
2	Literature Survey	Review of existing systems and related research in student performance prediction
3	System Analysis	Discussion of existing system, drawbacks, proposed system, advantages, applications, and requirements
4	System Architecture	The conceptual model that defines the structure, behavior, and views of a complex system.
5	System Design	Description of system architecture including UML diagrams and Data Flow Diagrams
6	Implementation	Details of system development and technologies used
7	Testing	Explanation of testing methods and validation of system performance
8	Result	Presentation of results, screenshots, performance analysis, and interpretation
9	Conclusion	Summary of the project and possible future improvements
10	Reference	List of sources and materials used
11	Appendix	Additional documents such as project files and QR code

Table: 1.4

CHAPTER 2

LITERATURE SURVEY

The literature survey discusses the use of machine learning in educational data analysis and student performance prediction. Various researchers have explored classification models, feature engineering techniques, and recommendation systems for improving educational outcomes.

Machine learning models such as Decision Tree, Logistic Regression, Support Vector Machine, Naive Bayes, and Random Forest are commonly used in educational prediction systems. Among these, Random Forest performs well because it handles multiple features effectively and reduces overfitting.

Recommendation systems are also widely used in education to suggest learning resources, course paths, and improvement strategies based on student behavior. Web-based dashboards and visual analytics are important for presenting prediction outcomes in an easy-to-understand format.

Topic	Author/ Publisher	Year of Publication	Advantages	Disadvantages
[1] A Machine Learning Approach for Predicting Academic Performance and Risk Levels	International Journal of Innovative Research in Technology (IJIRT)	2026	Uses Random Forest Regressor for accurate prediction; categorizes students into risk levels; provides personalized study plans and real-time intervention.	Depends heavily on quality and availability of student data; limited discussion of model scalability across institutions.
[2] Student Performance Prediction Using Machine Learning	International Journal of Creative Research Thoughts (IJCRT)	2025	Achieves high accuracy using XGBoost and Random Forest; supports early identification of at-risk students; improves academic outcomes through intervention.	Focuses mainly on historical academic data; may not consider all behavioral and socio-economic factors.
[3] Prediction of Student Academic Performance in Higher Education Using Machine Learning Techniques	International Journal for Multidisciplinary Research (IJFMR)	2025	Utilizes multiple ML algorithms; identifies key influencing factors such as attendance and previous grades; provides a data-driven prediction framework.	Performance may vary across datasets; limited discussion on real-time deployment and implementation challenges.
[4] Research on Student Achievement Prediction Method Based on Machine Learning	IEEE Xplore	2023	Provides an early warning system for failing students; achieves 88.78% prediction accuracy; assists teachers in targeted interventions.	Relies mainly on previous semester scores; may overlook lifestyle and behavioral factors affecting performance.
[5] A Systematic Literature Review of Student Performance Prediction Using Machine Learning Techniques	MDPI	2021	Comprehensive review of research trends; identifies effective machine learning techniques and data sources; useful for future researchers.	Literature review only; does not propose or implement a new predictive model.
[6] Predicting Students Academic Performance using an Improved Random Forest Classifier	IEEE Xplore	2020	Improved Random Forest classifier achieves better accuracy than traditional methods; effective in identifying at-risk students.	Focuses on algorithm improvement only; lacks personalized recommendation or intervention mechanisms.

Table: 2.1

CHAPTER 3

SYSTEM ANALYSIS

System analysis is the process of studying the existing methods, identifying limitations, and designing an improved solution. In the proposed project, the analysis focuses on how student performance can be predicted automatically and how recommendations can be generated using machine learning.

3.1 EXISTING SYSTEM

In the existing system, student performance is usually evaluated manually by teachers based on exam marks, attendance, and classroom observation. This approach is simple but not efficient when the number of students increases. It also does not analyze multiple performance-related factors simultaneously.

3.1.1 DRAWBACKS

- It depends mostly on manual evaluation.
- It takes more time and effort.
- It cannot analyze all factors effectively.
- It does not provide prediction in advance.
- It does not offer personalized recommendations.
- It is less efficient for large student datasets.
- It may contain human errors.
- It does not provide dashboards or automated reports.

3.2 PROPOSED SYSTEM

The proposed system is a machine learning-based web application that predicts student performance and provides recommendations. It uses academic, behavioral, and lifestyle-based features to classify students into performance levels.

The application includes separate roles for Student, Teacher, and Admin, along with dashboards, prediction history, and visual analytics.

3.2.1 ADVANTAGES

- Fast and automated prediction.
- Accurate classification using machine learning.
- Personalized academic recommendations.
- Secure role-based login system.
- Prediction history storage.
- Interactive dashboards.
- Visual analytics and charts.
- Better academic monitoring.
- Easy-to-use web interface.

3.3 APPLICATIONS

The system can be used in:

- Schools
- Colleges
- Coaching centers
- Student counseling systems
- Academic monitoring platforms
- Institutional dashboards

CHAPTER 4

SOFTWARE REQUIREMENT SPECIFICATIONS

4.1 Overview

The Software Requirement Specification defines the hardware, software, functional, and non-functional requirements needed to develop and run the system.

4.2 HARDWARE REQUIREMENTS

S.No	Component	Specification
1	Processor	Intel i3 or higher
2	RAM	4 GB minimum
3	Storage	500 GB or more
4	Monitor	Standard display
5	Keyboard	Standard keyboard
6	Mouse	Optical mouse
7	Internet	Required for deployment and updates

Table 4.2: Hardware Requirements

4.1 SOFTWARE REQUIREMENTS

S.No	Software Component	Specification
1	Operating System	Windows 10/11
2	Programming Language	Python
3	Backend Framework	Flask
4	Frontend	HTML, CSS, JavaScript
5	Database	SQLite
6	Machine Learning Library	Scikit-Learn
7	Data Processing	Pandas, NumPy
8	Visualization	Chart.js
9	IDE	VS Code

Table 4.3: Software Requirements

4.2 FUNCTIONAL REQUIREMENTS

S.No	Requirement	Description
1	Role selection	User chooses Student, Teacher, or Admin
2	Authentication	Secure login and logout
3	Prediction input	Student enters academic and lifestyle data
4	Performance prediction	System predicts student category
5	Recommendation generation	System provides improvement suggestions
6	Prediction history	System stores past predictions
7	Student dashboard	Displays personal results and charts
8	Teacher dashboard	Displays student analytics
9	Admin dashboard	Displays system-wide statistics
10	Report generation	Results can be exported or displayed clearly

Table 4.4:Functional Requirements

4.3 NON-FUNCTIONAL REQUIREMENTS

S.No	Requirement	Description
1	Performance	Fast prediction response
2	Reliability	Consistent results
3	Security	Role-based authentication
4	Scalability	Can support more users in future
5	Usability	Easy and clean interface
6	Maintainability	Easy to update and debug
7	Availability	Accessible through browser
8	Portability	Can run on multiple systems

Table 4.5: Non -Functional Requirements

CHAPTER 5

SYSTEM DESIGN

System Design is the process of defining the architecture, modules, interfaces, and data flow of the proposed system. The Student Performance Prediction and Recommendation System is designed as a web-based application that integrates Machine Learning with Flask to predict academic performance and provide personalized recommendations.

The system consists of multiple modules such as Authentication Module, Prediction Module, Recommendation Module, Dashboard Module, Teacher Monitoring Module, Admin Management Module, and Database Module. These modules work together to provide an efficient and intelligent platform for student performance analysis.

5.1 INPUT DESIGN

The input design describes the data collected from the user through the prediction form.

Input Field	Description	Type
Study Hours	Number of hours studied per day	Numeric
Attendance	Attendance percentage	Numeric
Sleep Hours	Average sleep duration	Numeric
Social Media Usage	Daily social media usage	Numeric
Previous Score	Previous academic score	Numeric
Part-Time Job	Yes/No	Categorical
Internet Quality	Internet accessibility	Categorical
Parent Education Level	Educational background of parents	Categorical
Extracurricular Activities	Participation status	Categorical
Mental Health	Mental well-being level	Numeric

Table 5.1: Input Design

Input Validation

- Mandatory fields must not be empty.
- Numeric values must be within valid range.
- Categorical inputs must be correctly selected.
- Attendance should not exceed 100.
- Study hours should not be negative.

5.2 PROCESSING DESIGN

The processing design explains how the user input is converted into a prediction.

1. The student enters academic and lifestyle-related details.
2. The system validates the input.
3. Categorical values are encoded into numerical form.
4. Numerical features are scaled using the saved scaler.
5. Engineered features such as discipline score and study efficiency are computed.
6. The processed data is passed to the trained Random Forest model.
7. The model predicts the student performance category.
8. The system calculates confidence score and risk level.
9. Recommendations are generated.
10. The result is stored in the database and displayed on the dashboard.

5.3 OUTPUT DESIGN

The output design shows how prediction results are presented to the user.

Output Components

- Predicted performance category.
- Confidence score.
- Risk level.
- Personalized recommendations.
- Graphical analytics.
- Prediction history.

5.4 SYSTEM ARCHITECTURE

The system architecture follows a layered approach.

Presentation Layer

- HTML pages
- CSS styling
- JavaScript interaction
- Student/Teacher/Admin dashboards

Application Layer

- Flask backend
- Authentication module
- Prediction module
- Recommendation module
- Dashboard logic

Data Layer

- SQLite database
- Prediction records
- User accounts
- ML model files (.pkl)

5.5 Workflow of the System

11. User opens the web application.
12. User selects role.
13. User logs in using valid credentials.
14. Student enters prediction details.
15. System validates and preprocesses input.
16. ML model predicts performance.
17. Recommendation engine provides suggestions.
18. Result is stored in database.
19. Dashboard displays prediction and analytics.

5.6 UML DIAGRAMS

5.6.1 USE CASE DIAGRAM

A Use Case Diagram represents the interaction between users and the Student Performance Prediction System. It shows how different actors communicate with the system and perform operations.

Actors:

- Student
- Teacher
- Admin

Use cases:

- Register
- Login
- Predict Performance
- View Recommendations
- View Prediction History
- View Student Dashboard
- View Teacher Dashboard
- View Admin Dashboard
- Logout

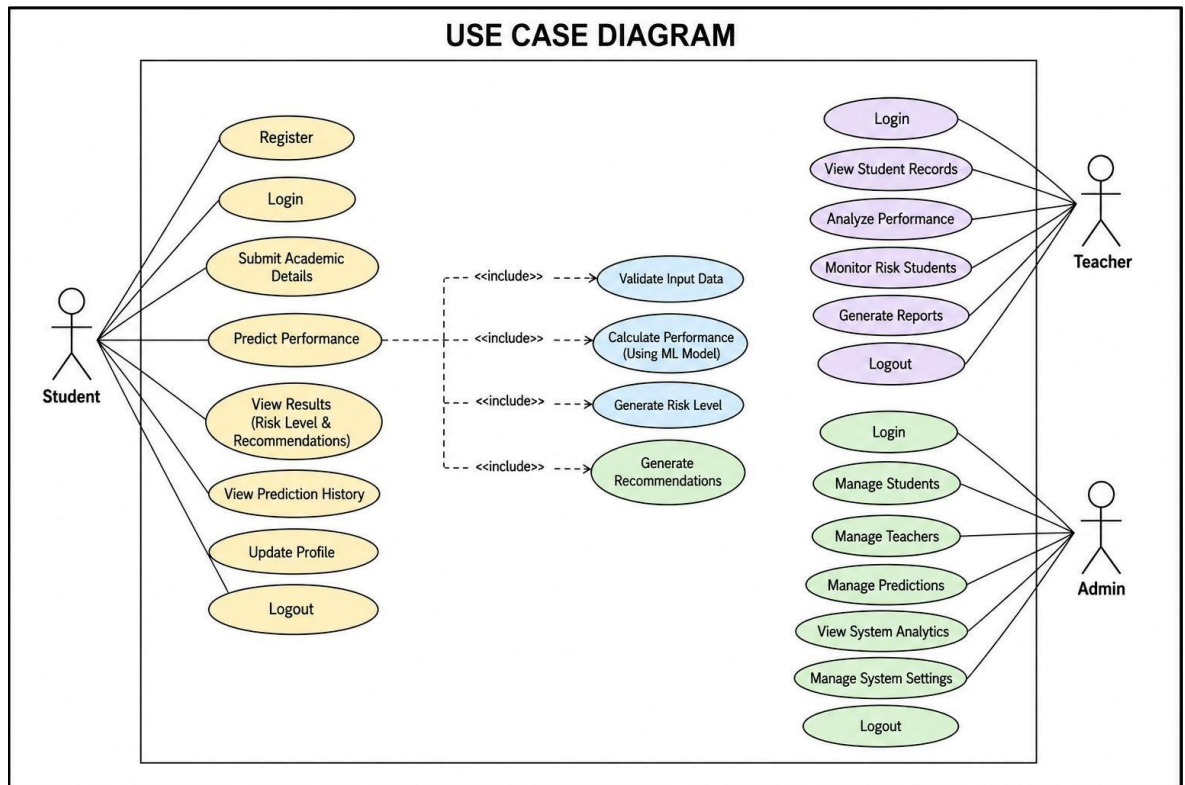


Fig 5.6.1: Use Case Diagram

5.6.2 CLASS DIAGRAM

The Class Diagram represents the internal structure of the Student Performance Prediction System. It shows classes, attributes, methods, and relationships among system components.

Main classes:

- User
- Prediction
- Student
- Teacher
- Admin
- ML Model
- Recommendation Engine
- Dashboard

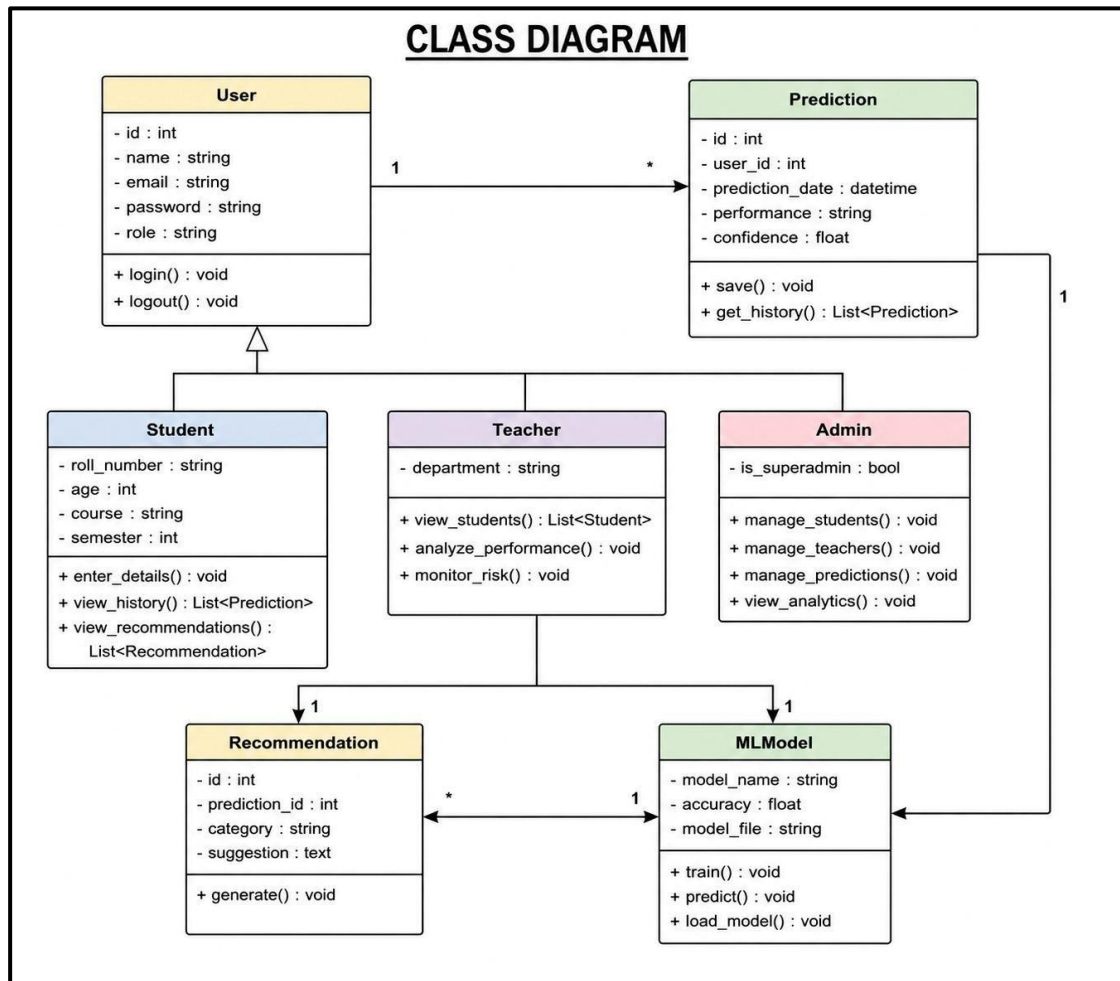


Fig 5.6.2: Class Diagram

5.6.3 SEQUENCE DIAGRAM

Flow:

- User logs in.
- User enters input.
- Flask validates request.
- Model predicts output.
- Recommendation engine generates suggestions.
- Result is displayed.

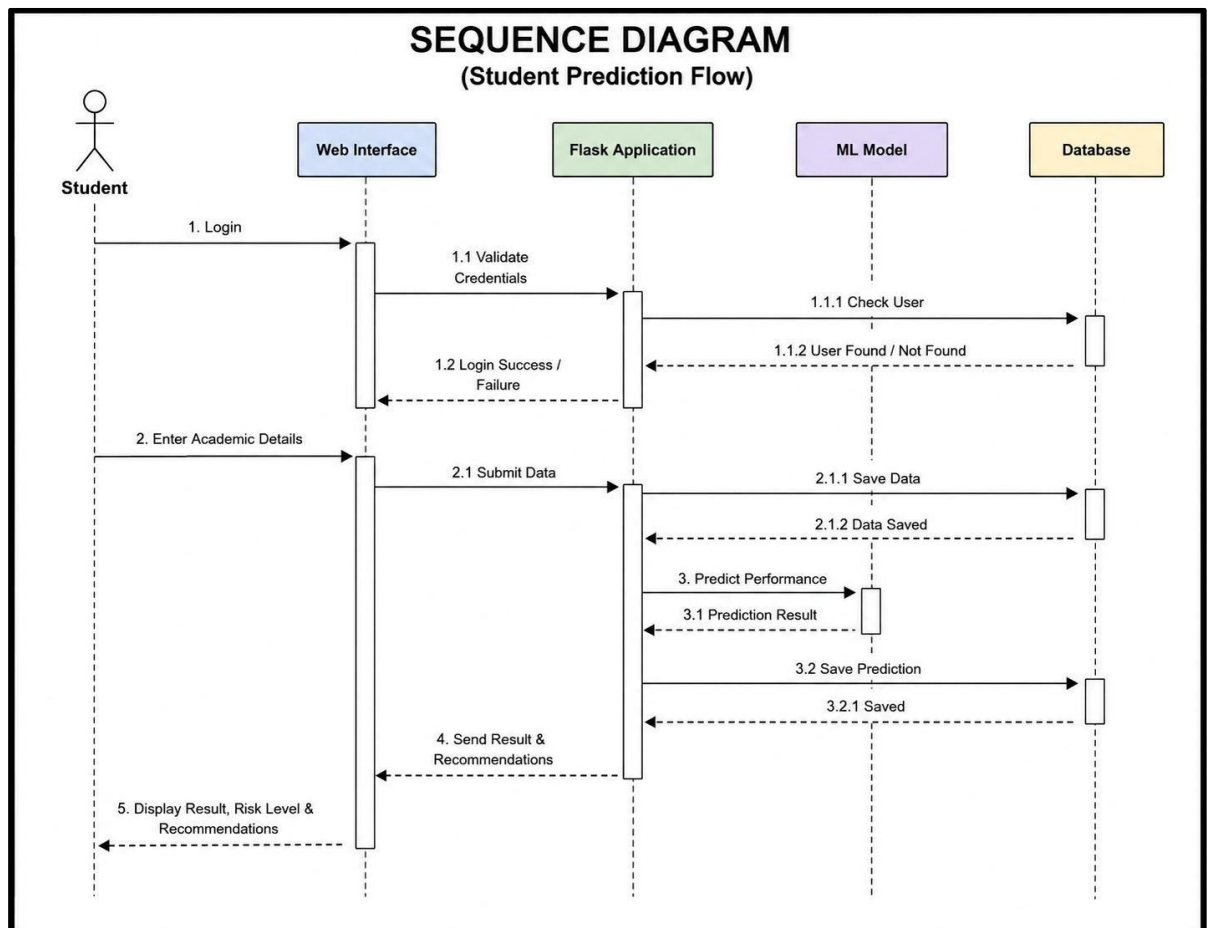


Fig 5.6.3: Sequence Diagram

5.6.4 ACTIVITY DIAGRAM

Flow:

- Login
- Enter details
- Validate data
- Predict performance
- Generate recommendations
- Save result
- Display dashboard

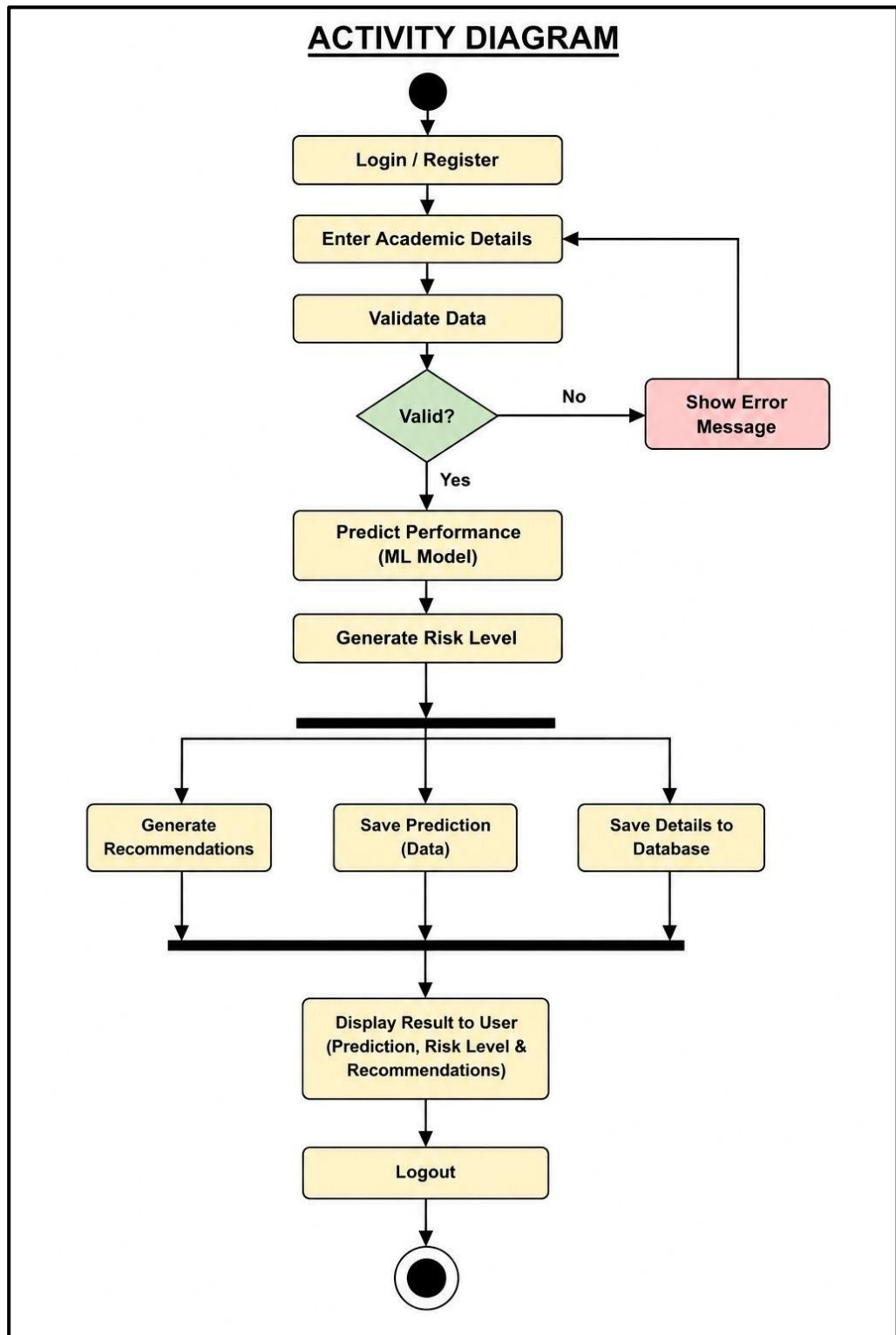


Fig 5.6.4: Activity Diagram

5.6.5 COMPONENT DIAGRAM

Components:

- User Interface
- Authentication Module
- Prediction Module
- Recommendation Engine
- Dashboard Module
- Database Module

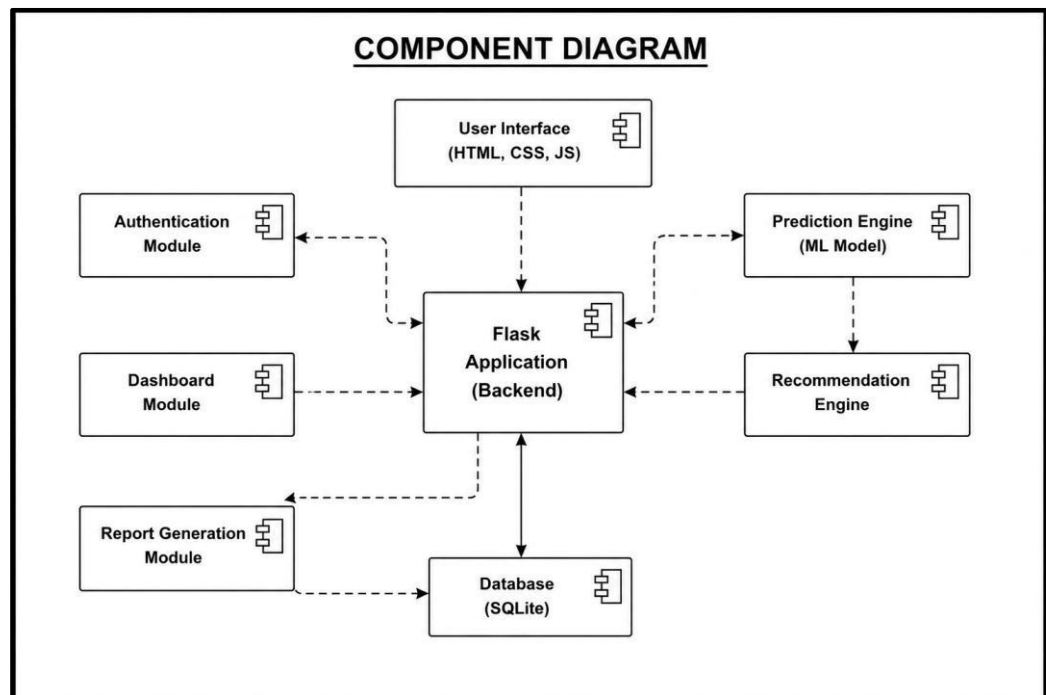


Fig 5.6.5: Component Diagram

5.6.6 DEPLOYMENT DIAGRAM

Deployment nodes:

- Client Browser
- Flask Web Server
- ML Model Service
- SQLite Database

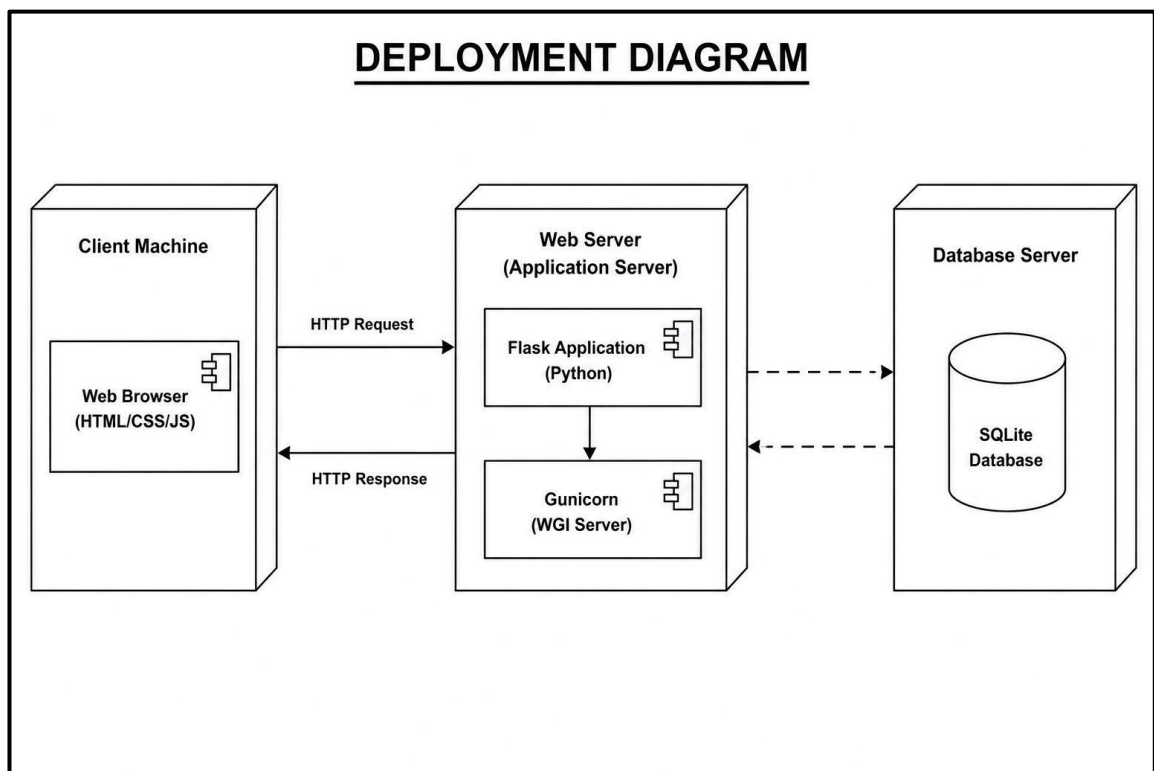


Fig 5.6.6: Deployment Diagram

5.7 Data Flow Diagram

5.7.1 Level 0 DFD

External Entity: User

Process: Student Performance Prediction System

Output: Prediction result and recommendation

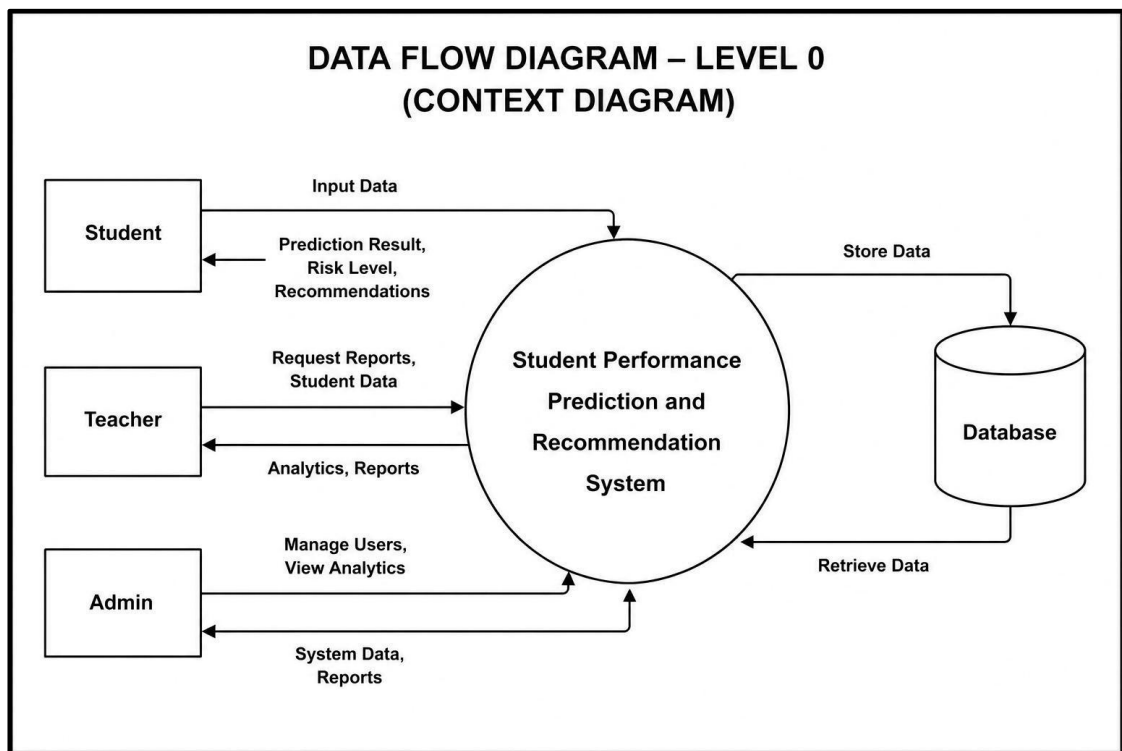


Fig 5.7.1: Data flow diagram

5.7.2 Level 1 DFD

Processes:

- User Login
- Data Collection
- Data Preprocessing
- ML Prediction
- Recommendation Generation
- Dashboard Display

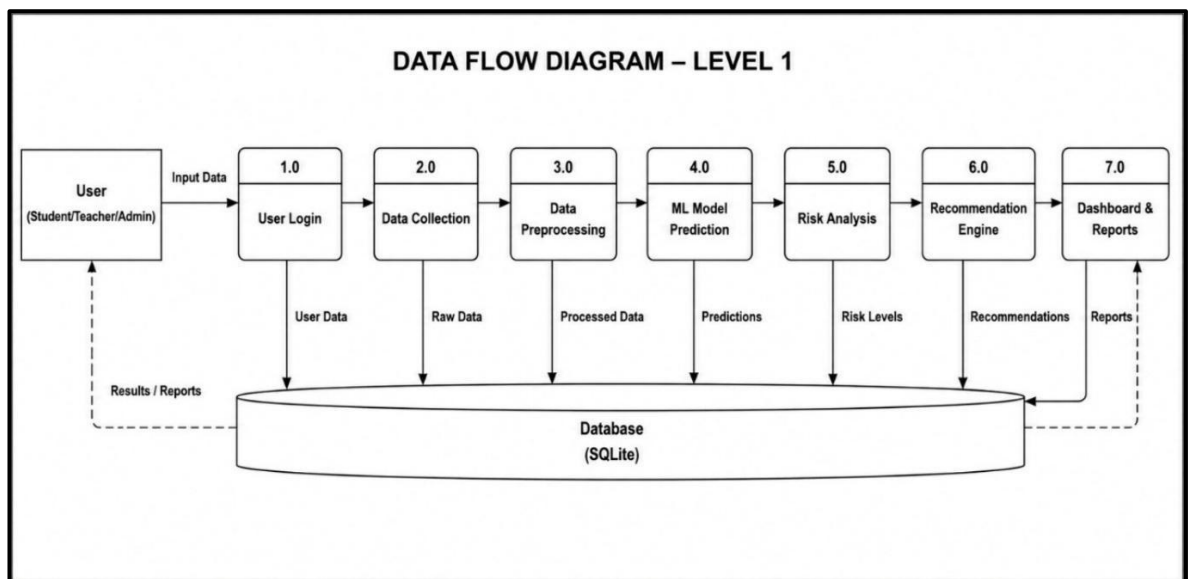


Fig 5.7.1: Data flow diagram

CHAPTER 6

IMPLEMENTATION

Implementation is the process of converting the system design into a fully functional software application. The Smart Student Performance Predictor and Recommendation System is implemented as a web-based application that combines Machine Learning techniques with educational analytics to predict student academic performance and provide personalized recommendations.

The system is developed using Flask as the backend framework and utilizes a Random Forest Classifier model trained on student academic and behavioral data. The application allows Students, Teachers, and Administrators to access different functionalities through role-based dashboards. The system analyzes various academic and lifestyle factors such as study hours, attendance, sleep duration, social media usage, parental education, and internet access to predict student performance levels.

The implementation focuses on providing an easy-to-use interface, accurate prediction mechanism, personalized recommendations, and graphical visualization of student performance data. The system ensures efficient data processing, secure authentication, and real-time prediction generation.

6.1 TECHNOLOGIES USED

The Smart Student Performance Predictor and Recommendation System is developed using modern web development and machine learning technologies. The technologies used in the project are listed below:

S.No	Technology	Purpose
1	Python	Programming Language
2	Flask	Web Application Framework
3	Scikit-Learn	Machine Learning Model Development
4	Pandas	Data Processing and Analysis
5	NumPy	Numerical Computations
6	HTML	Web Page Structure
7	CSS	User Interface Design
8	JavaScript	Interactive Web Features
9	Chart.js	Data Visualization and Graphs
10	SQLite	Database Management
11	Jupyter Notebook	Model Training and Testing
12	Visual Studio Code	Development Environment

Table 6.1: Tools And Technologies Used

6.2 SYSTEM ARCHITECTURE

The Smart Student Performance Prediction and Recommendation System follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer. The Presentation Layer provides an interactive web interface through which Students, Teachers, and Administrators interact with the system. It includes the home page, role selection page, login pages, prediction form, dashboards, and visualization pages. The Application Layer handles user authentication, data preprocessing, machine learning prediction, recommendation generation, and dashboard management. The Data Layer manages the SQLite database and trained machine learning model files, storing user information, prediction history, recommendations, and system records. This architecture ensures secure data management, efficient processing, accurate prediction generation, and smooth communication between all system components.

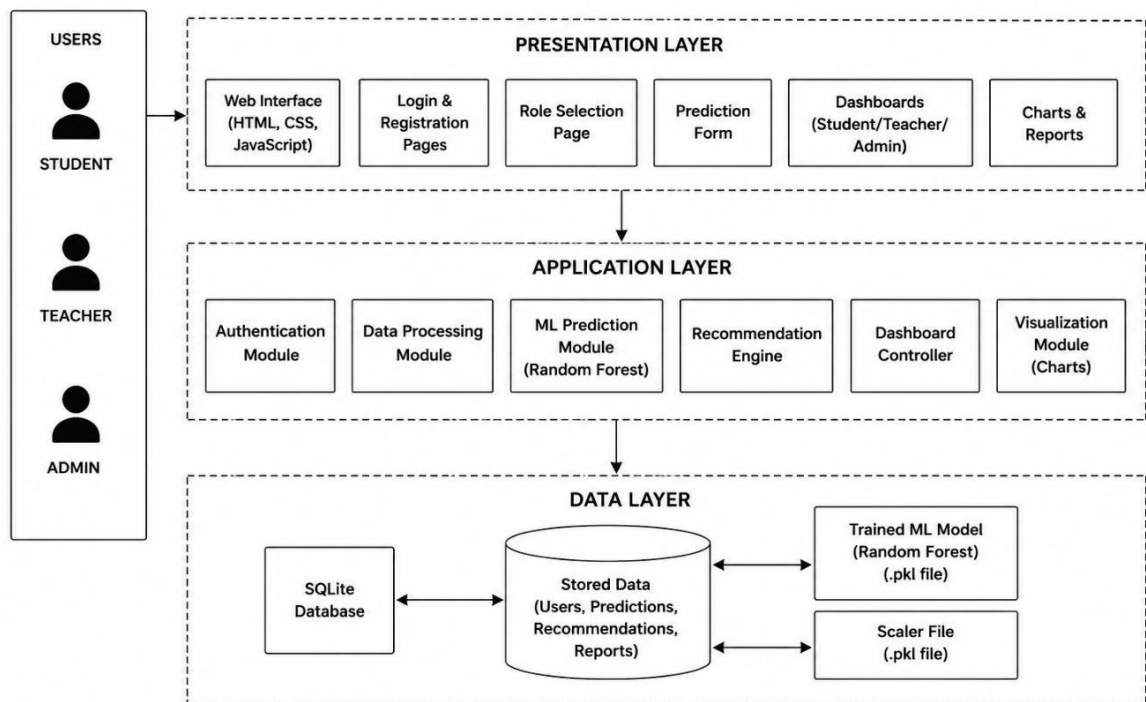


Fig 6.2.1 System Architecture

6.3 MODULE DESCRIPTION

The Smart Student Performance Prediction and Recommendation System is divided into several modules, each responsible for performing a specific task. These modules work together to provide accurate prediction results, personalized recommendations, and dashboard analytics.

1. Role Selection Module

This module allows users to select their role before accessing the system. The available roles are Student, Teacher, and Admin. Based on the selected role, the user is redirected to the corresponding login page.

Functions

- Role Selection
- User Redirection
- Access Control

Output

- Student Login Page
- Teacher Login Page
- Admin Login Page

2. Authentication Module

This module manages user authentication and authorization. It validates login credentials and grants access according to the user role.

Functions

- User Login
- User Logout
- Session Management
- Credential Validation

Output

- Secure Access
- Dashboard Redirection

3. Student Prediction Module

This is the core module of the system. It collects student academic and lifestyle-related information and predicts academic performance using the trained Random Forest model.

Input Parameters

- Study Hours
- Attendance Percentage
- Sleep Hours
- Previous Scores
- Social Media Usage
- Internet Access
- Parent Education Level
- Extracurricular Activities
- Part-Time Job Status

Output

- Performance Category
- Confidence Score
- Risk Level

4. Recommendation Module

This module generates personalized recommendations based on the prediction result and student input factors.

Functions

- Weakness Identification
- Improvement Suggestion Generation
- Academic Guidance

Output

- Personalized Recommendations
- Improvement Strategies

5. Student Dashboard Module

This module provides students with a complete overview of their prediction results and performance history.

Features

- Prediction History
- Performance Analytics
- Recommendation Display
- Charts and Graphs
- Risk Analysis

6. Teacher Dashboard Module

This module helps teachers monitor student performance and identify students who require academic support.

Features

- Student Performance Monitoring
- Risk Category Analysis
- Performance Distribution
- Dashboard Analytics

7. Admin Dashboard Module

This module provides administrative control over the system and allows monitoring of overall activities.

Features

- User Management
- Prediction Statistics
- System Analytics
- Performance Monitoring

8. Visualization Module

This module displays prediction results and performance trends using graphical representations.

Features

- Pie Charts
- Bar Charts
- Performance Distribution Graphs
- Risk Analysis Graphs

6.4 WORKING PROCEDURE

The Smart Student Performance Prediction and Recommendation System follows a systematic workflow to predict student performance and provide recommendations.

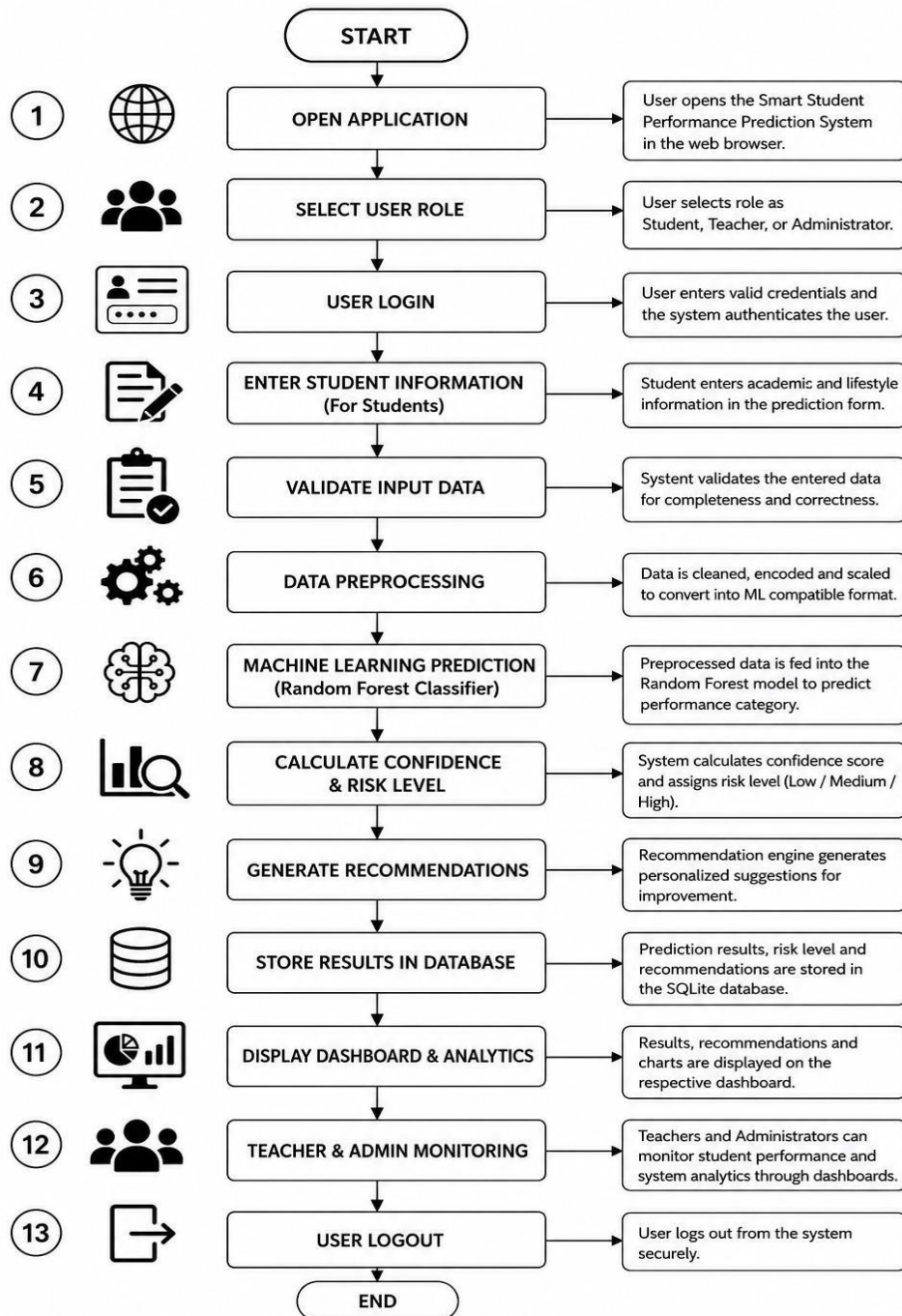


Fig 6.4.1: Working procedure

CHAPTER 7

TESTING

Testing is an essential phase in software development that ensures the system functions correctly according to the specified requirements. The main objective of testing is to identify errors, validate functionality, improve reliability, and ensure that the developed system performs efficiently under different conditions.

In the proposed Student Performance Prediction and Recommendation System, testing was performed on all major modules including authentication, prediction, recommendation generation, dashboard analytics, database operations, and user management.

The system was tested using different test cases to verify accuracy, security, usability, and performance.

7.1 TEST CASES

Test Case ID	Input	Expected Output	Actual Output	Status
TC01	Open application	Home page loads	Home page loaded	Pass
TC02	Select student role	Student login page	Student login page displayed	Pass
TC03	Enter valid login details	Login successful	Login successful	Pass
TC04	Enter prediction data	Prediction generated	Prediction generated	Pass
TC05	Leave mandatory field empty	Validation error	Error shown	Pass
TC06	Enter valid academic data	Recommendation displayed	Recommendation displayed	Pass
TC07	View dashboard	Prediction history visible	History displayed	Pass
TC08	Teacher login	Teacher dashboard opens	Dashboard opened	Pass
TC09	Admin login	Admin dashboard opens	Dashboard opened	Pass
TC10	Logout	Session closed	Logged out	Pass

Table 7.1: Test Cases

7.2 TESTING METHODS

Unit Testing

Each module was tested individually to ensure correct functionality.

Integration Testing

The interaction between prediction, recommendation, dashboard, and database modules was tested.

System Testing

The full application was tested as a complete system.

Validation Testing

Input fields were validated to ensure correctness and reliability.

7.3 RESULTS OF TESTING

All major modules functioned correctly. The prediction module generated outputs without error, the dashboards displayed the correct data, and the authentication system worked as expected. The application successfully handled validation errors and stored prediction history in the database.

CHAPTER 8

RESULTS

8.1 PRESENT RESULTS OBTAINED

The proposed Student Performance Prediction and Recommendation System was successfully developed and tested using machine learning and Flask technologies. The system accepts academic and behavioral input from the student and predicts the likely performance category. Based on the prediction output, the recommendation engine provides personalized suggestions to improve academic performance.

The system supports role-based access for students, teachers, and administrators. Students can enter prediction data, view the result, track their previous predictions, and receive improvement suggestions. Teachers can monitor student performance trends and identify students requiring support. Administrators can view overall system statistics and manage users through a centralized dashboard.

The model calculates a confidence score and risk level for every prediction. The risk level helps in identifying students who require immediate attention, while the confidence score reflects how strongly the model supports the predicted category.

The system also stores prediction results in the database for future reference. Charts and analytics improve usability and help users understand student performance patterns more clearly.

8.2 SCREENSHOTS

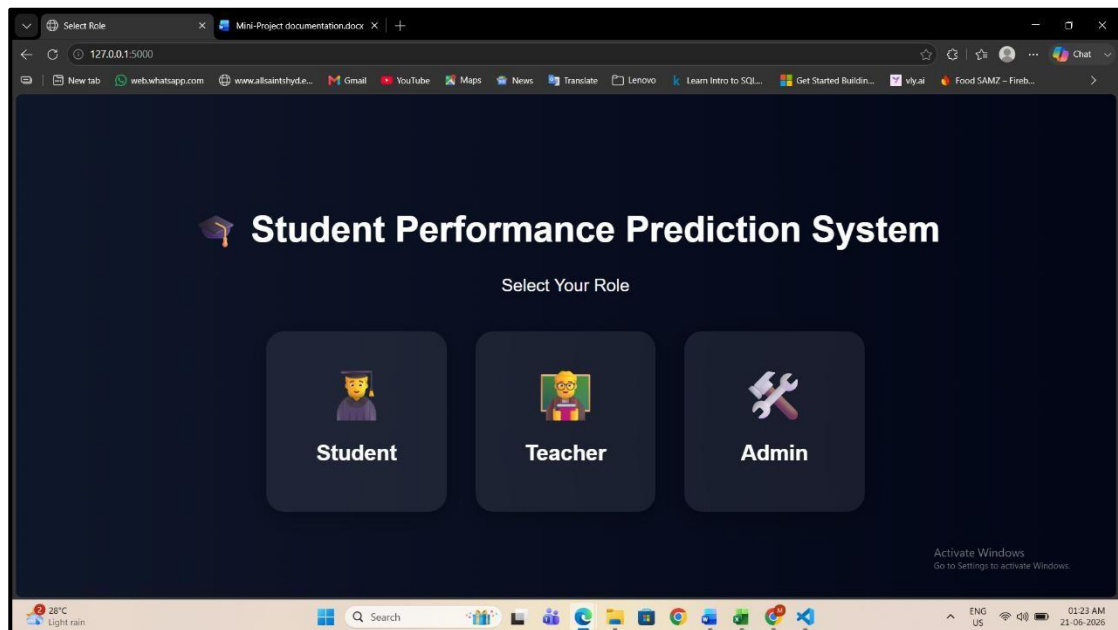


Fig 8.2.1.1: home page

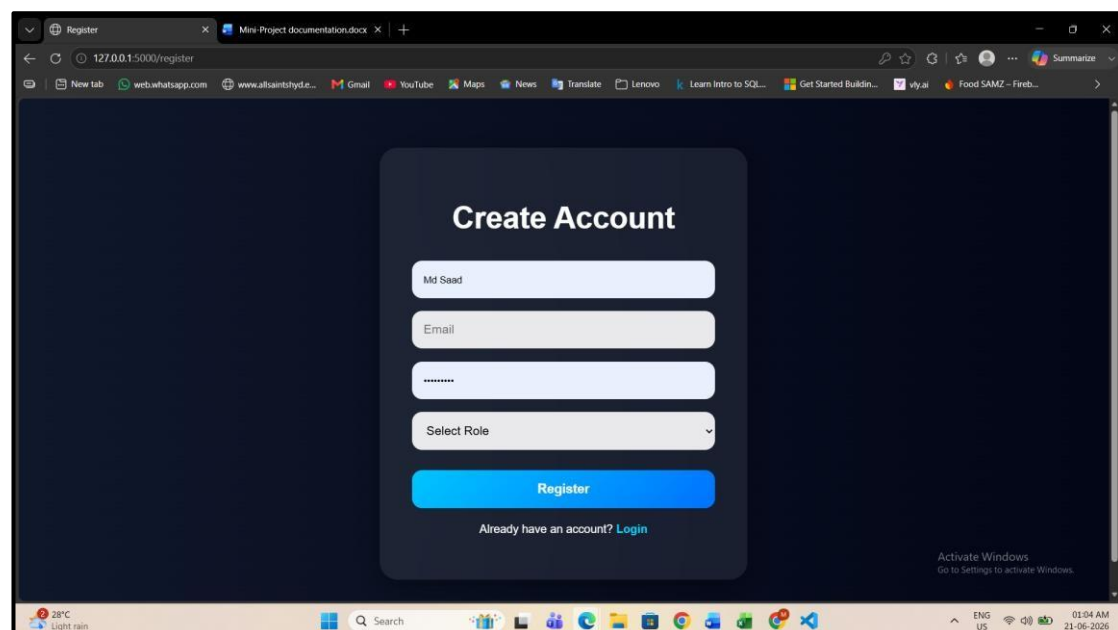
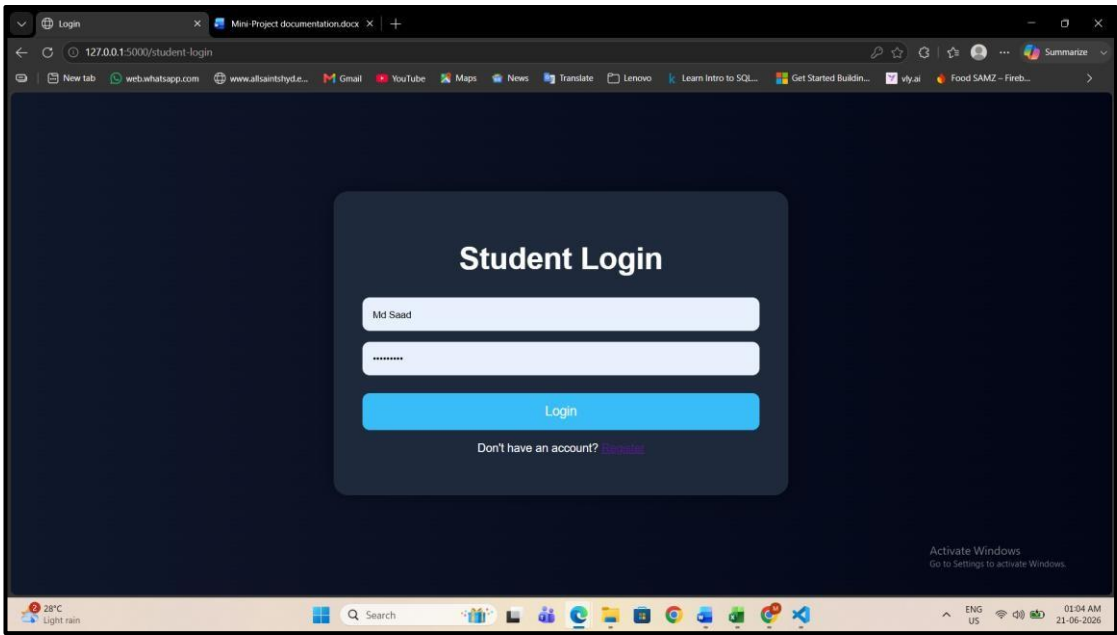


Fig 8.2.1.2: Role selection page

8.2.2 STUDENT DASHBOARD



[OBJ]

Fig 8.2.2.1: Student login page

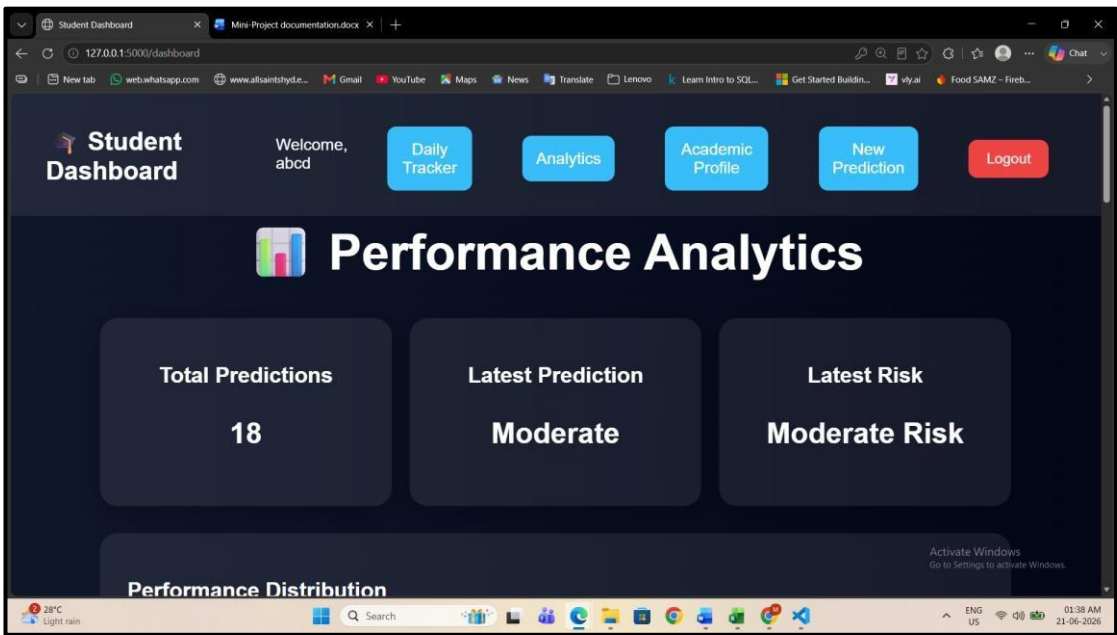


Fig 8.2.2.2: Performance Analytics

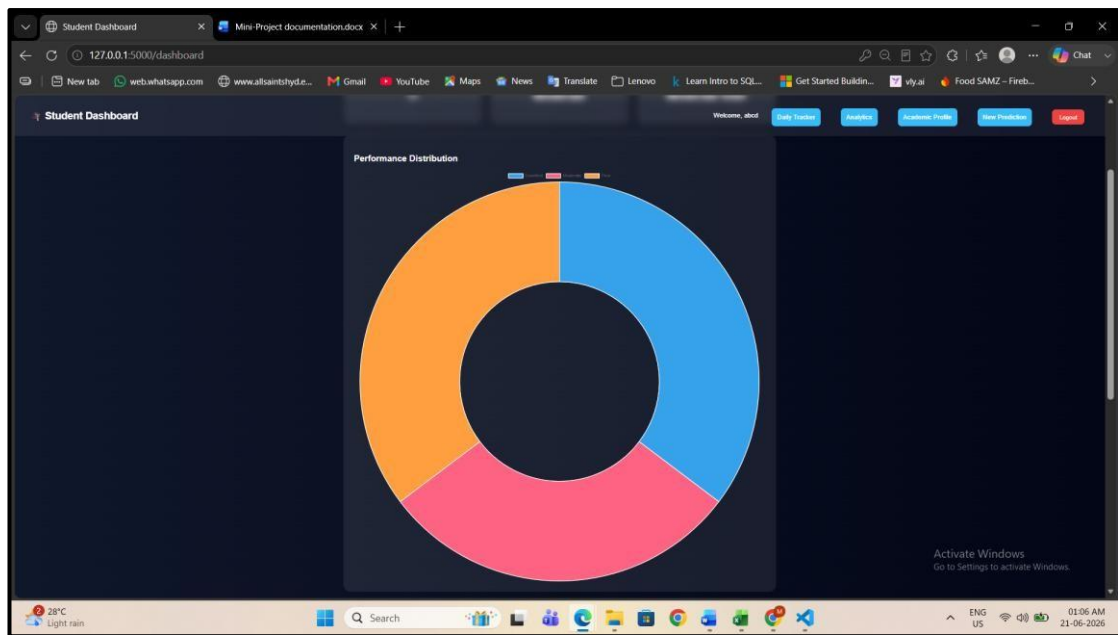


Fig 8.2.2.2: Performance Analytics

The screenshot shows the "Student Dashboard" with the "Prediction History" section active. The table below lists the prediction history with columns for Performance, Confidence, and Risk Level. The risk levels are indicated by colored buttons: orange for Moderate Risk, red for High Risk, and green for Low Risk.

Performance	Confidence	Risk Level
Medium	80.23%	Moderate Risk
Low	71.19%	High Risk
Medium	86.32%	Moderate Risk
Low	81.71%	High Risk
Medium	79.58%	Moderate Risk
Excellent	82.16%	Low Risk
Moderate	70.15%	Moderate Risk
Moderate	66.52%	Moderate Risk
Poor	57.23%	High Risk
Excellent	82.36%	Low Risk

Fig 8.2.2.2: Performance Analytics

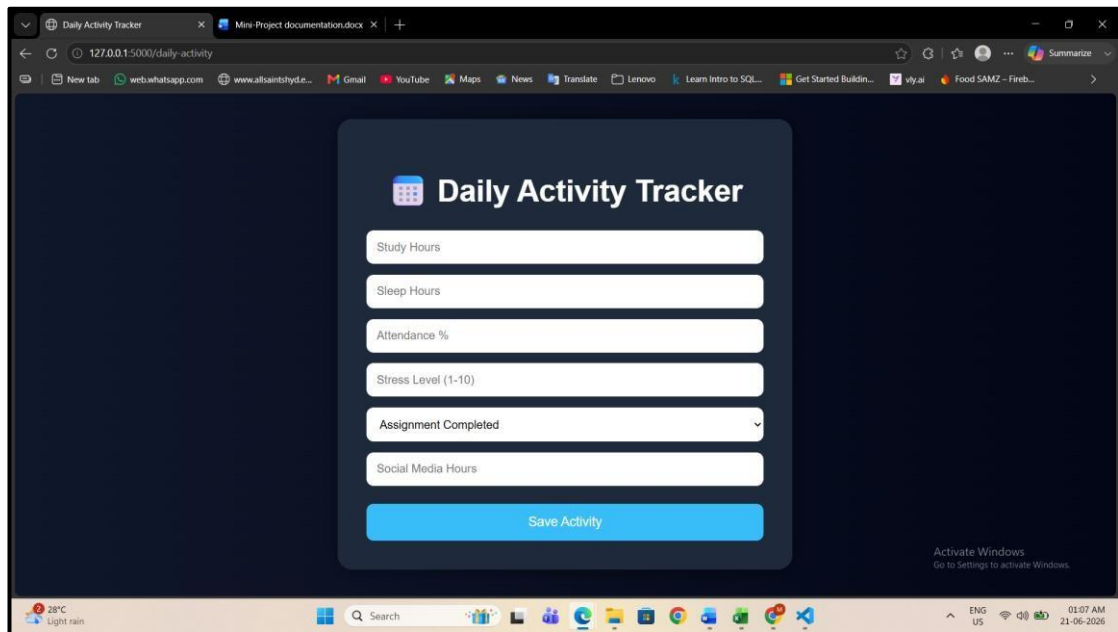


Fig 8.2.2.3: Daily Activity Tracker

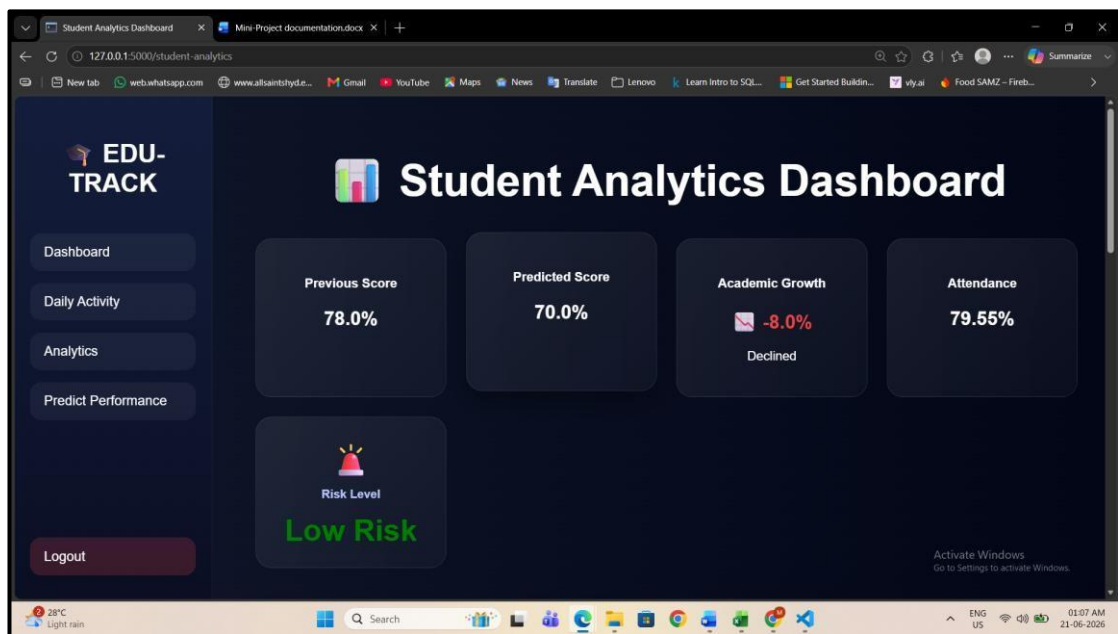


Fig 8.2.2.4: Student Analytics Dashboard

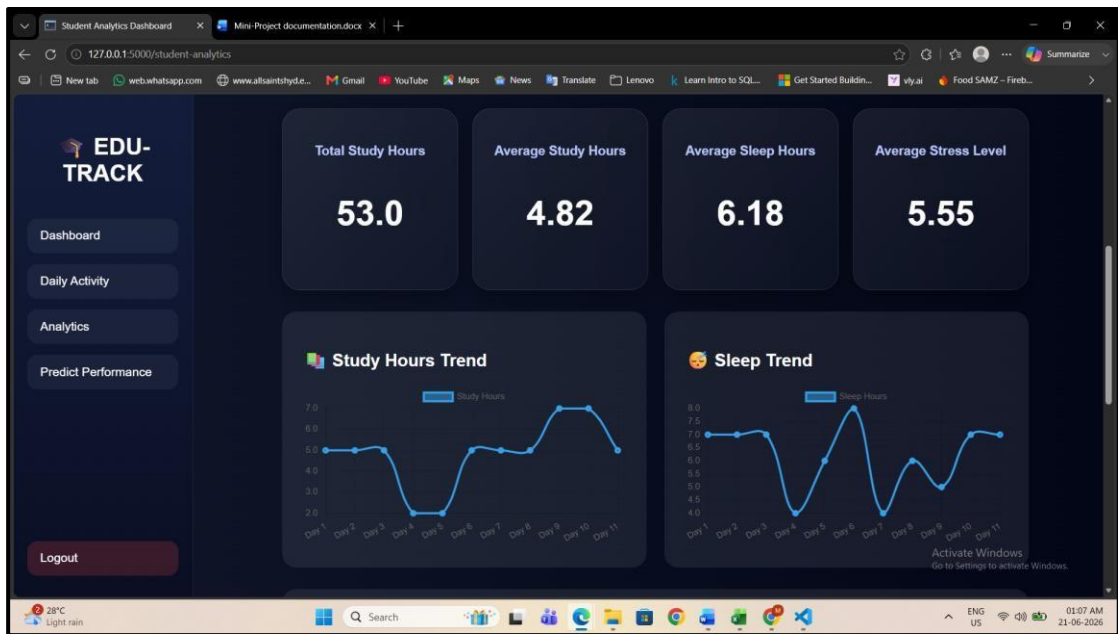


Fig 8.2.2.4: Student Analytics Dashboard

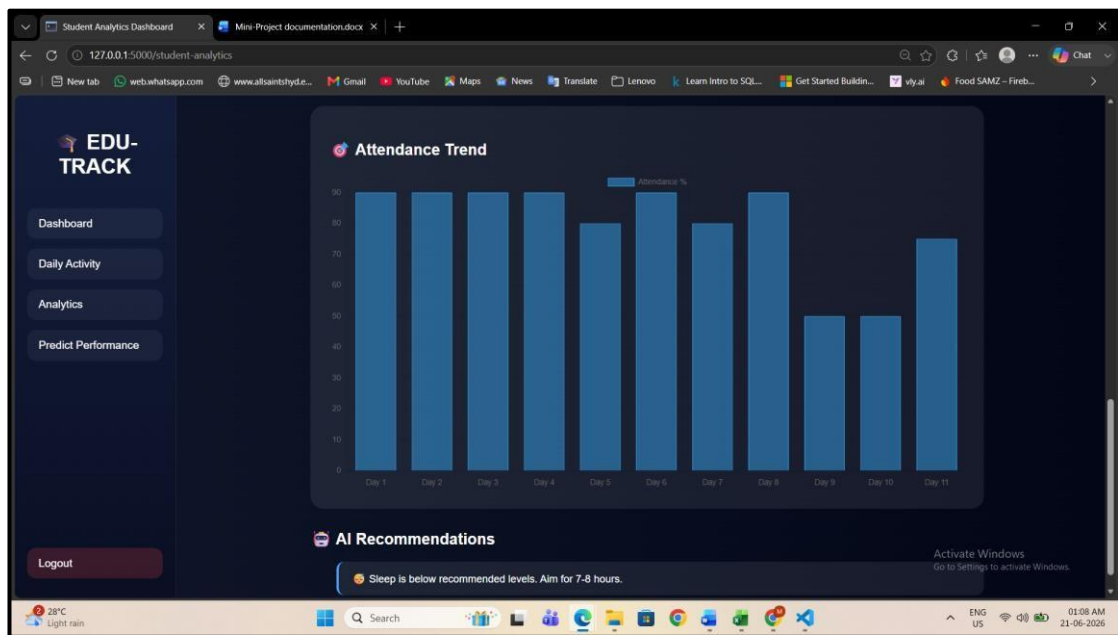


Fig 8.2.2.4: Student Analytics Dashboard

Student Profile
Complete your academic information

Roll Number
75

Semester
6

Previous Exam Score
78.0

Save Profile

Fig 8.2.2.5: Previous score input

Student AI System Welcome, abcd **Logout**

Student Prediction Form

Personal Information

Age

Select Gender

Study Details

Study Hours Per Day

Attendance Percentage

Sleep Hours

Digital Habits

Social Media Hours

Netflix Hours

Internet Quality

Fig 8.2.2.6: Student Information

Student AI System

Welcome, abcd [Logout](#)

Lifestyle

Additional Details

[Predict Performance](#)

Fig 8.2.2.6: Student Information

Student AI System

Welcome, abcd [Logout](#)

Student Performance Dashboard

Performance

Moderate

Confidence

55.71%

Risk Level

Moderate Risk

Important Factors

Study Hours
Attendance
Sleep Quality
Mental Health
Lifestyle Score
Discipline Score

Fig 8.2.2.7: Recommendation Dashboard

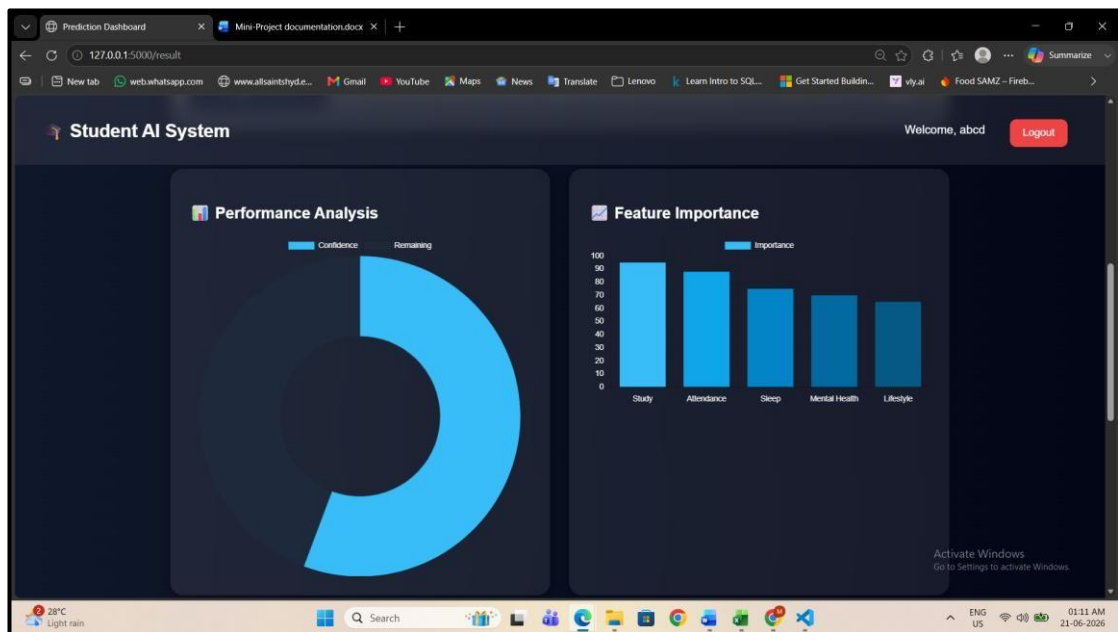


Fig 8.2.2.7: Recommendation Dashboard

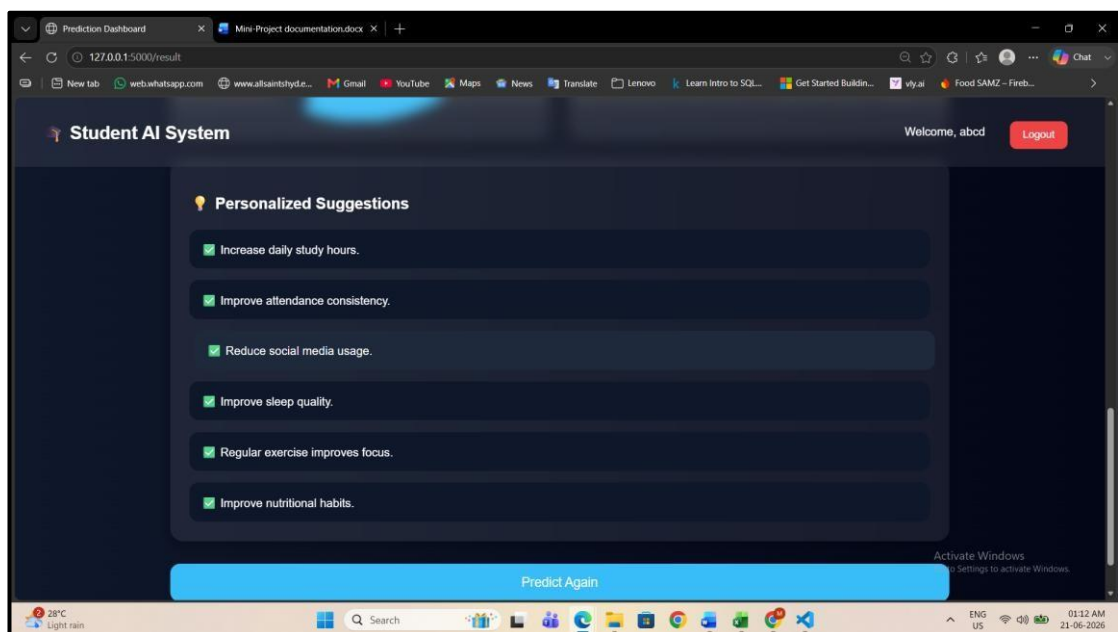


Fig 8.2.2.7: Recommendation Dashboard

8.2.3 TEACHER DASHBOARD

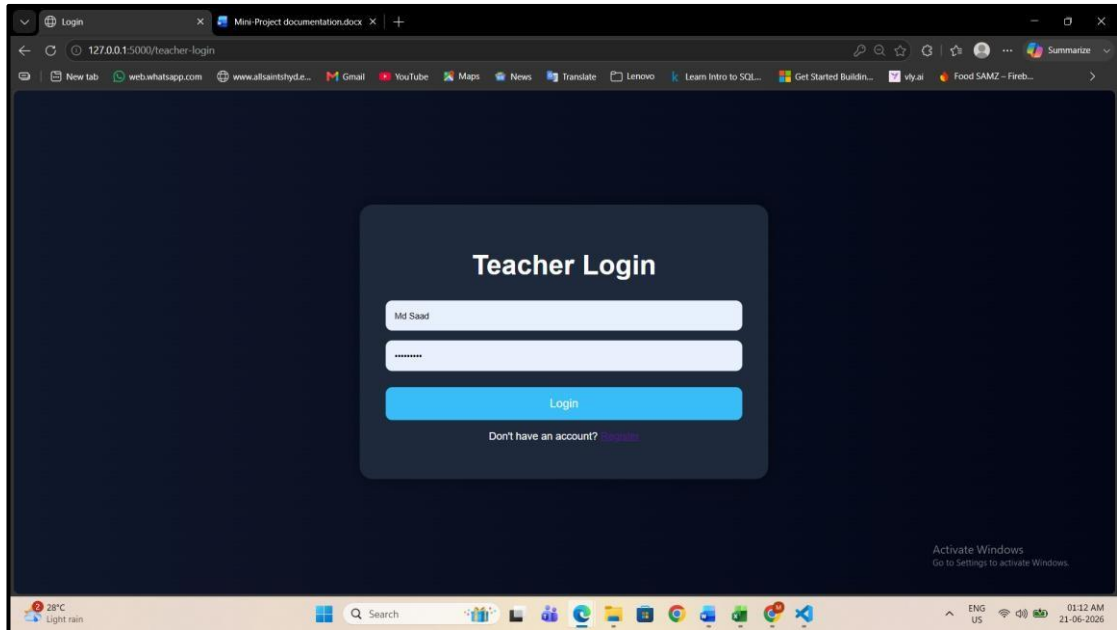


Fig 8.2.3.1 Teacher Login page

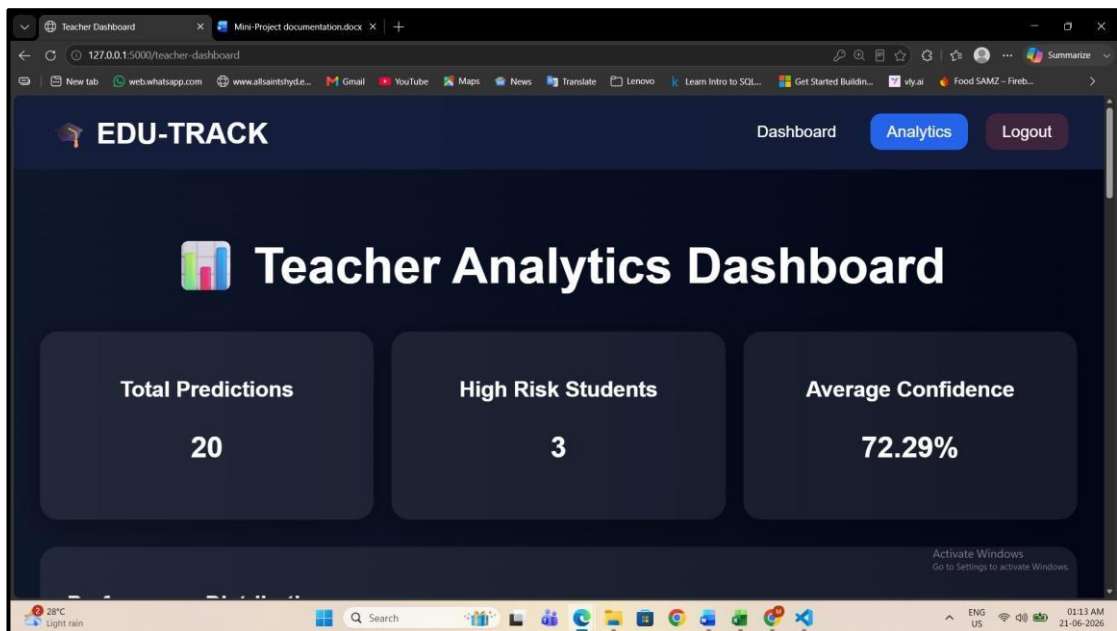


Fig 8.2.3.2: Teacher analytics Dashboard

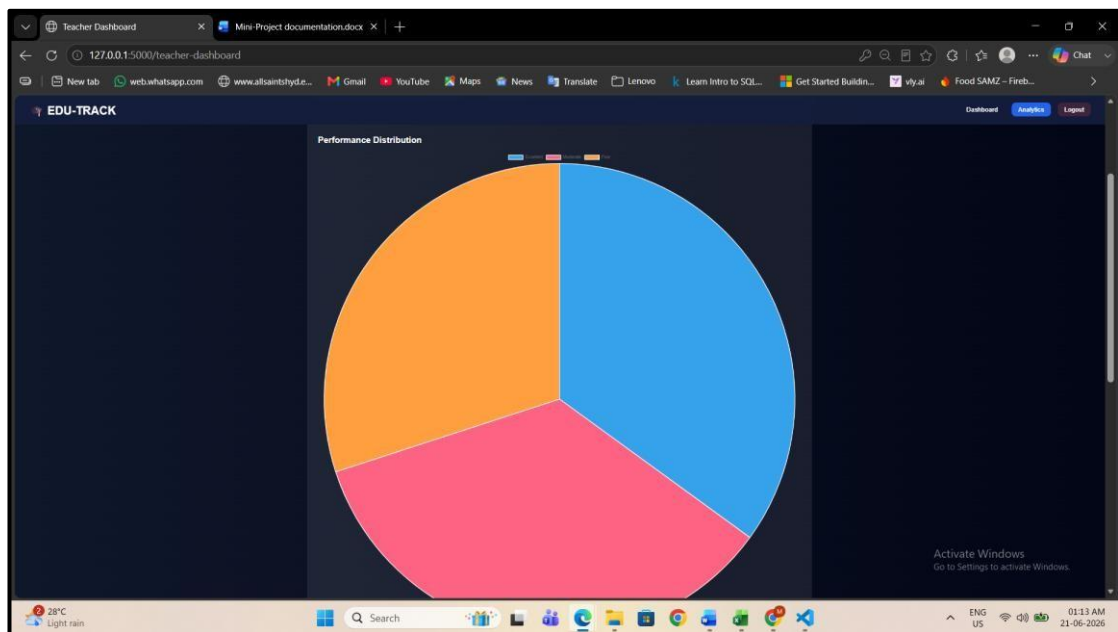


Fig 8.2.3.2: Teacher analytics Dashboard

The screenshot displays the 'Teacher Dashboard' for 'EDU-TRACK' with the 'Analytics' tab selected. The main section is titled 'All Student Predictions' and contains a table with four columns: Student, Performance, Confidence, and Risk. The table lists 10 student records, each with a unique ID (abcd), a performance level, a confidence percentage, and a risk assessment (Moderate Risk, High Risk, or Low Risk). The dashboard includes a top navigation bar with 'Dashboard', 'Analytics', and 'Logout' buttons. The bottom status bar shows the temperature as 28°C, weather as 'Light rain', and the time as 01:14 AM on 21-06-2026.

Student	Performance	Confidence	Risk
abcd	Medium	80.23%	Moderate Risk
abcd	Low	71.19%	High Risk
abcd	Medium	86.32%	Moderate Risk
abcd	Low	81.71%	High Risk
abcd	Medium	79.58%	Moderate Risk
abcd	Excellent	82.16%	Low Risk
abcd	Moderate	70.15%	Moderate Risk
abcd	Moderate	66.52%	Moderate Risk
abcd	Poor	57.23%	High Risk
abcd	Excellent	87.26%	Low Risk

Fig 8.2.3.2: Teacher analytics Dashboard

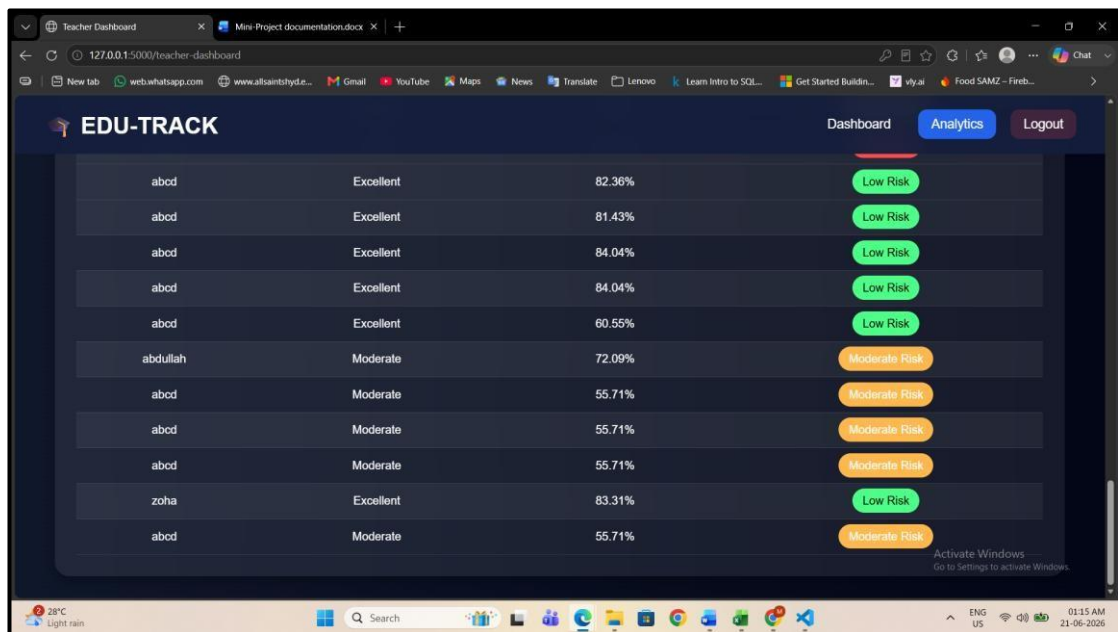


Fig 8.2.3.2: Teacher analytics Dashboard

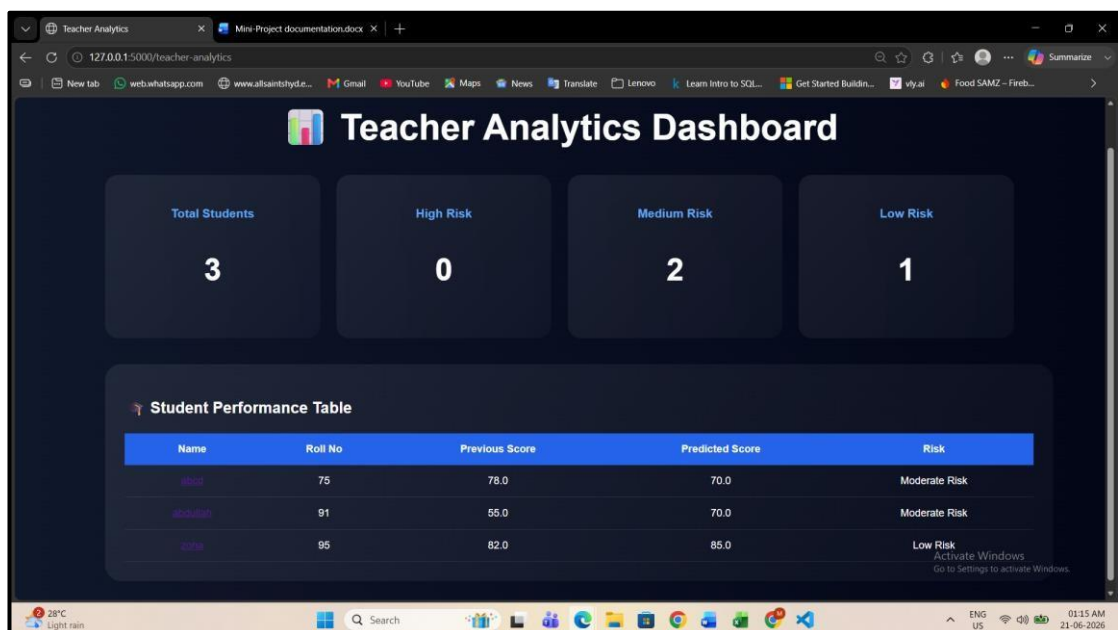


Fig 8.2.3.3: Teacher analytics Dashboard

8.2.4 ADMINSTRATOR DASHBOARD

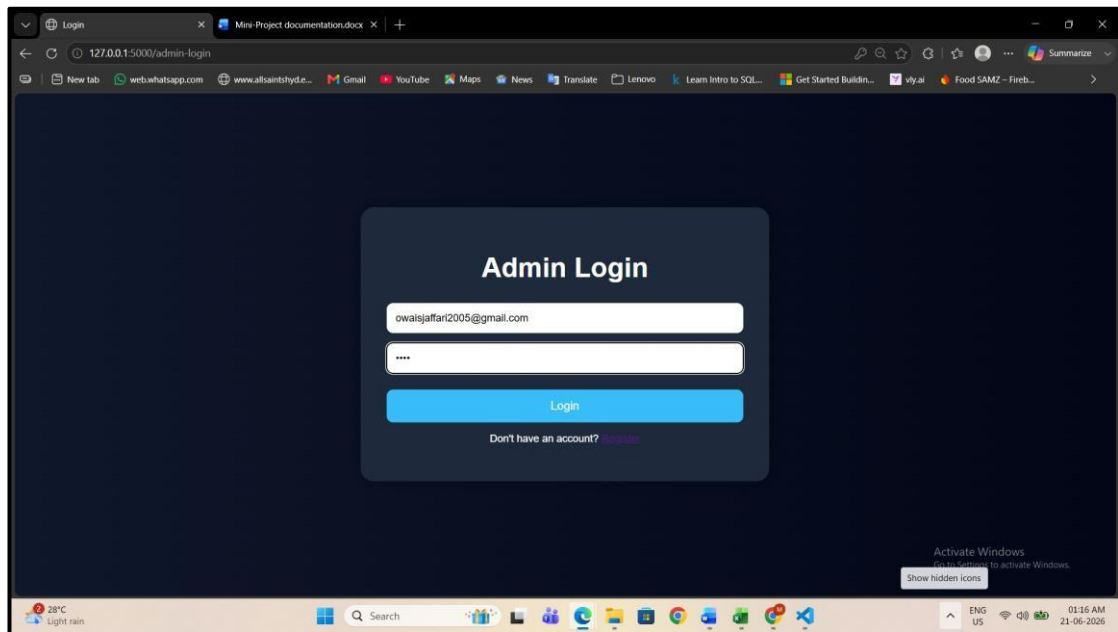


Fig 8.2.4.1: Admin Login Page

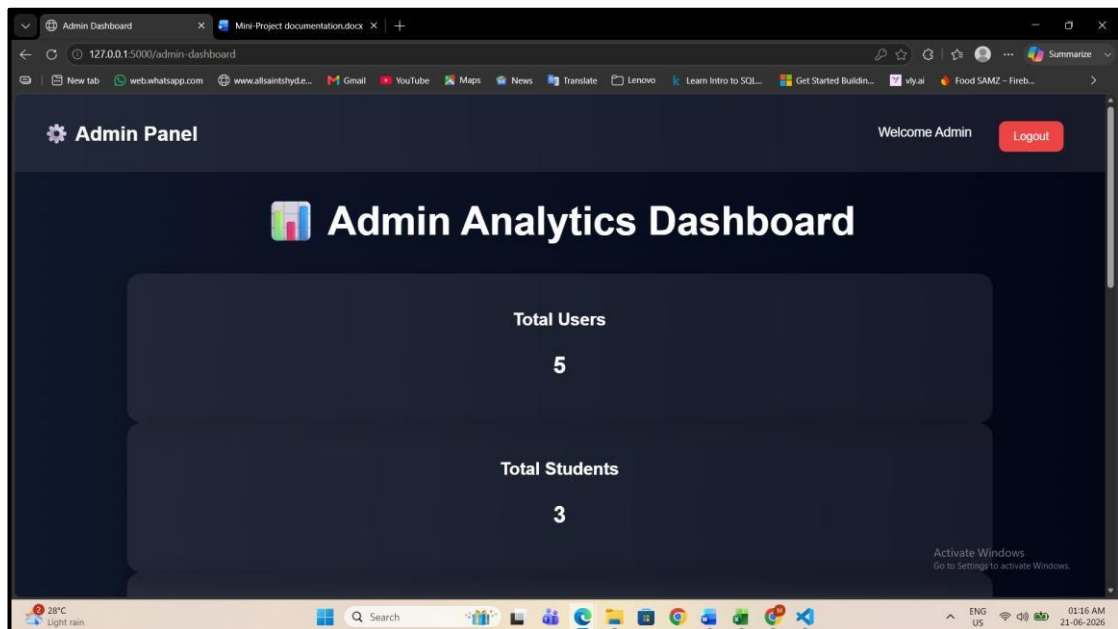


Fig 8.2.4.2: Admin Dashboard

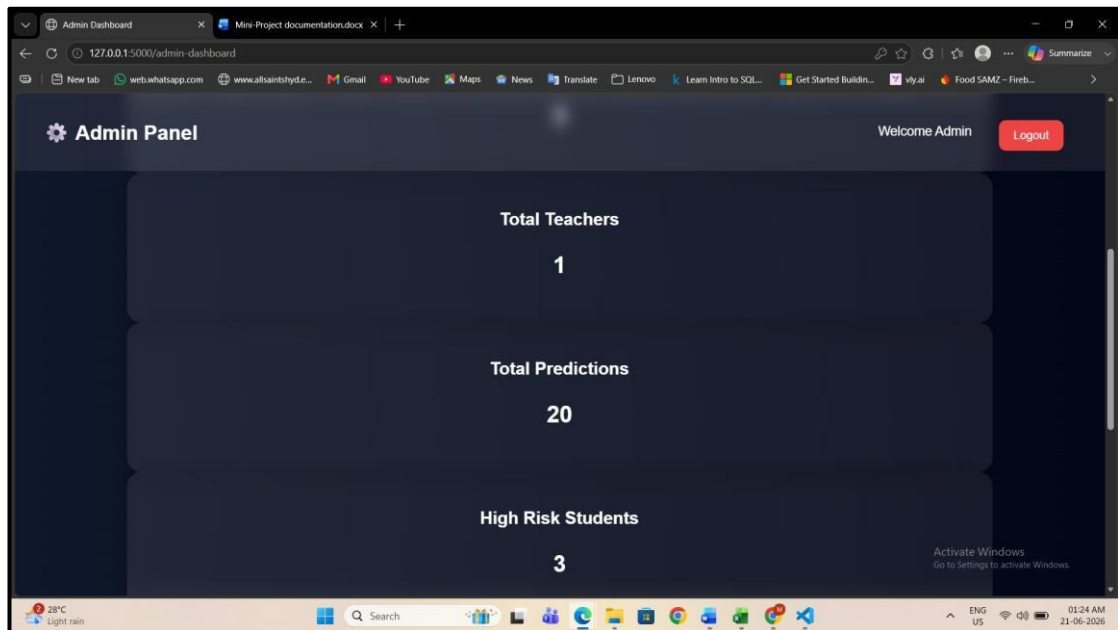


Fig 8.2.4.2: Admin Dashboard

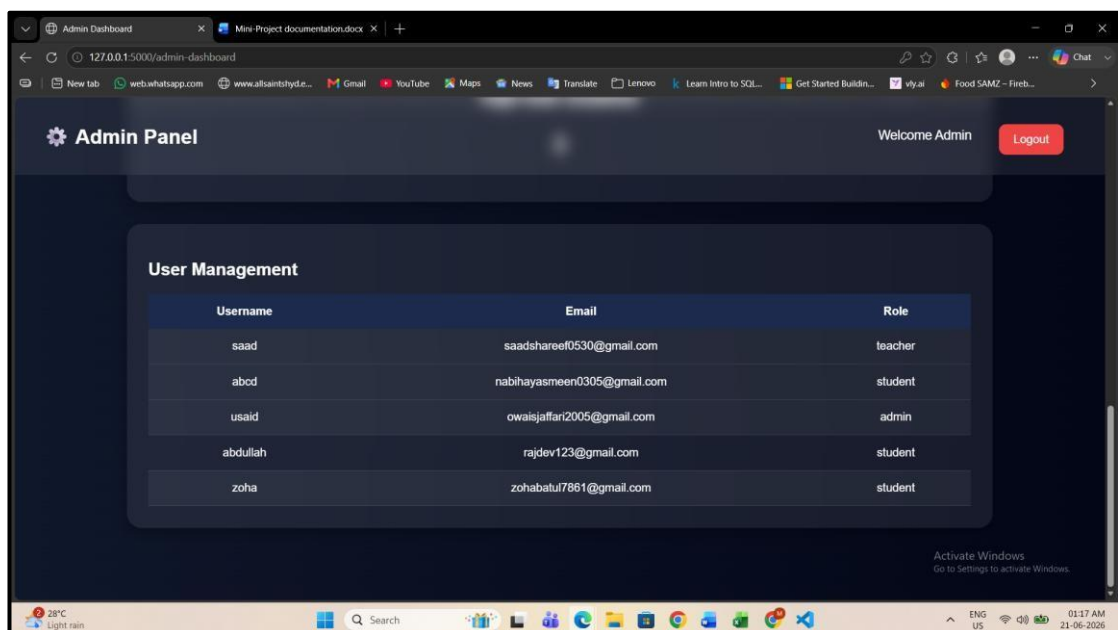


Fig 8.2.4.2: Admin Dashboard

8.3 PERFORMANCE ANALYSIS

The performance of the proposed Advanced Student Performance Prediction and Recommendation System was evaluated based on prediction capability, recommendation generation, dashboard functionality, system responsiveness, and overall user experience. The developed system successfully integrates Machine Learning algorithms with a web-based platform to provide accurate and meaningful insights regarding student academic performance.

The Random Forest Classifier model demonstrated effective prediction capabilities by analyzing multiple academic, behavioural, and lifestyle parameters. The model successfully classified students into different performance categories based on the information provided through the prediction form. The generated predictions were accompanied by confidence scores, allowing users to understand the certainty level of the prediction results.

The recommendation engine effectively analyzed the identified weaknesses of students and generated personalized suggestions for academic improvement. Students with low study hours received recommendations to increase study time, while students with poor attendance were advised to improve classroom participation. The system therefore not only predicts performance but also assists users in improving their future academic outcomes.

The role-based architecture enhanced system usability by providing separate dashboards for students, teachers, and administrators. Students were able to access their prediction history and recommendations, teachers could monitor student performance trends, and administrators could view overall system statistics and user activity.

The visualization module further improved system effectiveness through charts and graphical representations. Performance trends, prediction distributions, and risk-level analyses enabled users to interpret results more efficiently compared to traditional tabular reports.

8.4 INTERPRETATION OF RESULTS

Prediction Category	Interpretation
High Performance	Student demonstrates strong academic behavior and consistent performance
Moderate Performance	Student has potential for improvement through better study habits
Low Performance	Student requires academic guidance and targeted interventions
Low Risk	Student is likely to maintain good academic performance
Moderate Risk	Student should improve certain academic factors
High Risk	Student requires immediate attention and academic support

Table 8.1: Interpretation Of Results

CHAPTER 9

CONCLUSION

The Student Performance Predicting System Using Random Forest Algorithm was successfully designed and implemented as a web-based machine learning application for predicting students' academic performance and providing personalized recommendations for academic improvement. The system utilizes academic, behavioural, and lifestyle-related factors such as study hours, attendance, sleep duration, social media usage, previous academic scores, extracurricular participation, internet accessibility, and parental education level to analyze and predict student performance accurately.

The proposed system employs the Random Forest Classifier, which effectively classifies students into performance categories such as High, Medium, and Low. In addition to performance prediction, the system generates confidence scores, determines academic risk levels, and provides personalized recommendations that help students improve their study habits, learning strategies, and overall academic achievement.

The application was developed using Python, Flask, Scikit-Learn, SQLite, HTML, CSS, JavaScript, and Chart.js, providing an interactive and user-friendly environment. The inclusion of role-based access for Students, Teachers, and Administrators enables efficient performance monitoring, prediction history management, and educational analytics through dedicated dashboards and graphical visualizations.

The results obtained demonstrate that machine learning techniques can be effectively applied in the education domain to support data-driven decision-making and early identification of students who may require additional academic guidance. The system not only assists students in understanding their academic standing but also helps educators monitor student progress and implement timely interventions when necessary.

Overall, the developed system successfully achieves its objectives by combining machine learning, recommendation systems, and web technologies into a unified platform for student performance analysis. The project provides an efficient, reliable, and scalable solution that can contribute to improving academic outcomes and enhancing educational support systems in modern learning environments.

9.1 Future Enhancements

The proposed system can be further enhanced by incorporating advanced technologies and additional functionalities to improve prediction accuracy, usability, and scalability. Future enhancements include:

- Integration of Deep Learning algorithms for improved prediction performance.
- Development of a mobile application to provide easy access across multiple devices.
- Integration with institutional databases and Learning Management Systems (LMS) for real-time data collection.
- Implementation of an AI-powered chatbot to provide academic guidance and student support.
- Addition of email and notification services for alerting students, teachers, and parents regarding academic performance.
- Development of a Parent Dashboard for monitoring student progress and recommendations.
- Deployment on cloud platforms to support large-scale institutional usage.
- Implementation of Explainable Artificial Intelligence (XAI) techniques to improve transparency and interpretation of prediction results.
- Inclusion of advanced analytics and reporting features for institutional decision-making.
- Support for multilingual interfaces to improve accessibility for diverse users.

These enhancements can transform the system into a comprehensive educational intelligence platform capable of supporting academic performance monitoring, personalized learning, and data-driven educational management.

CHAPTER-10

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CHAPTER-11

APPENDIX

11.1 Project Documents QR Code



**QUALITY ANALYSIS AND ALIGNMENT OF STUDENT PROJECTS WITH PROGRAM
OUTCOMES (POs) AND SUSTAINABLE DEVELOPMENT GOALS (SDGs)**

Course Name:	MINI PROJECT	AY	2025-26
Course Code:	7PW663ML	Semester	VI
Name of the Guide:	Mrs K Unnati Mohan	Class	CSM - B
PROJECT TITLE			
Student Performance Predicting System using Random Forest Algorithm			
STUDENTS			
S. No	HT No	Name of the Student	
1.	160723748075	Mohammed Saad Shareef	
2.	160723748091	Abdullah Jaffari Syed	
3.	160723748095	Zoha Batul	

Course Outcome: After the successful completion of this course, the student will be able

CO No	Course Outcome	BTL	POs, PSOs Mapped
7PW663ML.1	Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to the real-world problems.	Apply	PO1, PO2, PO3, PO6, PO8, PO11, PSO1, PSO2, PSO3
7PW663ML.2	Analyze and formulate the problem, and design solution of the selected project.	Analyze	PO1, PO2, PO3, PO4, PO5, PO6, PO8, PO11, PSO1, PSO2, PSO3
7PW663ML.3	Evaluate different solutions based on economic and technical feasibility.	Evaluate	PO3, PO4, PO5, PO6, PO7, PO8, PO11, PSO1, PSO2, PSO3
7PW663ML.4	Effectively plan, develop and deploy a project and confidently perform all aspects of project management.	Create	PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3
7PW663ML.5	Demonstrate effective written and oral technical communication skills.	Create	PO2, PO5, PO7, PO9, PO10, PO12, PSO1, PSO2, PSO3

CO-PO-PSO MAPPING

3 – High, 2– Moderate, 1 – Slight

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
7PW663ML.1	3	2	2	-	-	3	-	2	-	-	2	-	3	3	3
7PW663ML.2	3	3	2	3	2	2	-	2	-	-	2	-	3	3	3
7PW663ML.3	-	-	3	3	2	2	2	2	-	-	3	-	3	2	2
7PW663ML.4	-	-	3	3	3	2	2	3	3	3	3	2	3	3	3
7PW663ML.5	-	2	-	-	2	-	2	2	3	3	2	2	2	2	2

CO-PO-PSO MAPPING (With Performance Indicators)

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	P S O 3
7PW663M L.1	1.1.2 1.2.1 1.3.1 1.4.1	2.1.2 2.1.3 2.2.3 2.3.1 2.4.1	3.1.1 3.1.3 3.1.4 3.1.5 3.2.1 3.2.3 3.3.1 3.3.2	-	-	6.1.1 6.2.1 6.3.1 6.3.2 6.4.1	-	8.2.1 8.2.2 8.2.3	-	-	11.1.2 11.2.1 11.2.2 11.3.1 11.3.2	-	High	High	High
7PW663M L.2	1.1.2 1.2.1 1.3.1 1.4.1	2.1.1 2.1.2 2.1.3 2.2.3 2.3.1 2.4.1 2.4.3 2.4.4	3.1.1 3.1.3 3.1.4 3.1.5 3.1.6 3.2.1 3.2.3 3.3.1 3.3.2 3.4.1 3.4.2	4.1.1 4.1.2 4.1.3 4.2.1 4.3.1 4.3.2 4.3.3 4.3.4	-	6.1.1 6.3.1 6.3.2 6.4.2	-	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	-	-	11.1.2 11.2.1 11.2.2 11.3.2	-	High	High	High
7PW663M L.3	-	2.2.2 2.2.3 2.2.4 2.3.2 2.4.2 2.4.3 2.4.4	3.1.5 3.2.1 3.2.2 3.2.3 3.3.1 3.3.2 3.4.1 3.4.2	4.1.2 4.2.1 4.2.2 4.3.1 4.3.2 4.3.3 4.3.4	5.2.1 5.2.2 5.3.1 5.3.2	6.1.1 6.2.1 6.3.1 6.3.2 6.4.2	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	-	10.1.1 10.1.2 10.2.1	11.1.2 11.2.1 11.2.2 11.3.2	-	High	Mod	Mod
7PW663M L.4	-	2.3.1 2.3.2 2.4.1 2.4.2 2.4.3 2.4.4	3.1.6 3.2.1 3.2.2 3.3.2 3.4.1 3.4.2	4.1.2 4.1.3 4.2.1 4.2.2 4.3.1 4.3.3	5.1.1 5.1.2 5.2.1 5.2.2 5.3.1 5.3.2	6.1.1 6.2.1 6.3.1 6.3.2 6.4.2	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	9.1.1 9.1.2 9.1.3 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1 10.3.1 10.3.2	11.1.1 11.1.2 11.2.1 11.2.2 11.3.2	12.1.1 12.1.2 12.2.1 12.2.2 12.3.1 12.3.2	High	High	High
7PW663M L.5	-	2.2.1 2.2.2 2.2.3 2.2.4	-	-	5.1.1 5.1.2 5.2.1 5.2.2 5.3.1 5.3.2	-	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	9.1.1 9.1.2 9.1.3 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1 10.3.1 10.3.2	11.1.1 11.1.2 11.2.1 11.2.2 11.3.2	12.1.1 12.1.2 12.2.1 12.2.2 12.3.1 12.3.2	Mod	Mod	Mod



Project Specific Learning Outcomes

LO#	Outcome Description
1	The system predicts student academic performance using Machine Learning techniques based on academic and behavioral factors.
2	It identifies students who are at risk of poor academic performance and provides early intervention insights.
3	The system generates personalized recommendations to help students improve their learning outcomes and study habits.
4	It provides role-based dashboards for students, teachers, and administrators to monitor performance effectively.
5	It offers an interactive web-based platform for prediction, visualization, and academic performance analysis.

PO -WK Mapping:

PO#	PO Description	Mapped Wks	PO#		Mapped Wks
PO1	Engineering Knowledge	WK1, WK2, WK3, WK4	PO7	Environment and sustainability	WK5, WK7
PO2	Problem Analysis	WK1, WK2, WK3, WK4	PO8	Ethics	WK7, WK9
PO3	Design / Development of Solutions	WK1, WK2, WK3, WK4, WK5, WK6, WK7	PO9	Individual & Team Work	WK9
PO4	Conduct Investigations of complex problems	WK2, WK3, WK4, WK8	PO10	Communication	WK7, WK9
PO5	Engineering Tool usage	WK2, WK6	PO11	Project Management & Finance	WK5, WK6, WK7
PO6	The Engineer and society	WK1, WK5, WK7	PO12	Life-long Learning	WK8

SDG Mapping:

SDG	Mapped Indicator	SDG	Mapped Indicator	SDG	Mapped Indicator
1 NO POVERTY		7 AFFORDABLE AND CLEAN ENERGY		13 CLIMATE ACTION	
2 ZERO HUNGER		8 DECENT WORK AND ECONOMIC GROWTH		14 LIFE BELOW WATER	
3 GOOD HEALTH AND WELL-BEING	3.4.1	9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	9.5.1	15 LIFE ON LAND	
4 QUALITY EDUCATION	4.1.2	10 REDUCED INEQUALITIES	10.3.1	16 PEACE, JUSTICE AND STRONG INSTITUTIONS	
5 GENDER EQUALITY		11 SUSTAINABLE CITIES AND COMMUNITIES		17 PARTNERSHIPS FOR THE GOALS	

QUALITY ANALYSIS

Project collaboration (Tick ✓ Whichever is applicable and present the details below):

Industry ✓	Research Lab	Professional Society	Community	Alumni	Regulating agency	Govt.	Other ✓
Details: The proposed system supports educational institutions by predicting student performance using Machine Learning techniques. It helps teachers identify students who require additional academic support and provides personalized recommendations to improve learning outcomes, thereby enhancing educational quality and student success.							

Project Achievements:

Does the Abstract (prepared/ reviewed / Approved) have the following stated outcomes mentioned?	Yes
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S. No	Description	Details
1.	Prototype development	Developed a functional web-based prediction system.
2.	Technology based solution developed	Implemented Machine Learning for performance prediction and recommendations.
3.	Publication	--
4.	Start-up incubation	--
5.	Competition prizes	--
6.	Community / Industry / HEI service	Supports students and educators in academic performance monitoring.
7.	SDG Indicators	3.4.1, 4.1.2, 9.5.1, 10.3.1
8.	Grants / Financial Aid	--
9.	Multi-disciplinary	Combines Artificial Intelligence, Machine Learning, Data Analytics, and Education Technology.
10.	Internship / Employment	Helps students improve academic performance through personalized recommendations and predictive insights.
11.	Self-employment skill	Enhances analytical thinking, decision-making, and problem-solving skills.
12.	Use of real-time data	Processes student academic and behavioural data for performance prediction.

Signature of the Guide

Signature of the Project Coordinator